

# The Semi-Humanoid Robot Market 2026

Global competitive landscape: wheeled, tracked and pedestal dual-arm humanoid robots

218 robots · ~200 companies · China · USA/Canada · Europe · Japan · Korea · Rest of world

Market study compiled 2 August 2026

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# 1 Executive summary

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**The semi-humanoid — a human-like dual-arm upper body on a wheeled, tracked or pedestal base — is where humanoid robotics is actually being deployed in 2026.** While bipedal humanoids dominate headlines and venture valuations, wheeled dual-arm robots dominate paying work: of the 218 systems documented in this study, 82 are shipping products and a further 31 are formally announced. China's "2026 Wheeled Humanoid Robot Industry Development Blue Book" counts >5,000 wheeled-humanoid units shipped in China in 2025 (~25–28% of Chinese humanoid volume), and Interact Analysis states plainly that wheeled platforms are "the preferred near-term form factor for real-world industrial deployment." Morgan Stanley's influential humanoid forecasts *exclude* wheeled-base torsos entirely — meaning the segment this study covers is systematically under-represented in headline market numbers, and correspondingly under-analyzed by competitors.

## Seven findings that shape the opportunity:

- **China leads on volume and velocity.** 63 of 218 documented robots are Chinese; AgiBot alone shipped 5,100+ units in 2025 and passed 15,000 cumulative robots by June 2026. Real factory orders exist (Fulin Precision ~100× A2-W; Longcheer's hundreds-of-millions-RMB G2 order), and the supplier layer (chassis, actuators, hands) is commoditizing fast — entry prices now reach ¥26,900 (Unitree R1-D) and ~\$1,000–1,500 (SwitchBot Onero H1 target).
- **The US pivoted from bipeds to economics.** The 2025–26 US wave (Sunday Memo, Weave Isaac at \$7,999, Dexterity Mech, Dyna, Walden Robotics at a \$1.1B valuation out of stealth with Toyota as investor-customer) sells labor economics — RaaS by the hour, home robots by subscription — not anthropomorphism.
- **Regulation just redrew the map.** FCC Public Notice DA 26-786 (28 July 2026) added foreign-produced "advanced robotic devices" to the US Covered List: new foreign-made mobile, networked, AI-enabled robots — wheeled semi-humanoids squarely included — lose US equipment authorization unless granted a conditional approval. This fences the largest single market against Chinese, Korean, Japanese *and* European vendors alike, and materially raises the value of domestic/allied manufacturing footprints.
- **Europe is strong in research, fragmented in product.** PAL Robotics, Pollen/ Hugging Face, Neura's MiPA (€9,999 announced, deliveries slipping), igus's €47,999 Iggy Rob, Humanoid UK's \$1.35B valuation and Duatic's ETH-bred Alpha coexist with the cautionary tales of Aldebaran/Pepper (liquidated 2025) and Cybedroid. German research depth (DLR, KIT, TUM, FZI, Fraunhofer lineage) is world-class and under-commercialized — a talent and IP pool a German entrant can draw on.
- **Incumbents are converging on exactly this form factor.** Toyota's ELEY (QDD joints, scapular axis, wheeled base), Yaskawa's MOTOMAN NEXT + its quiet ownership of Tokyo Robotics, Midea's six-armed MIRO U, Kawasaki/Foxconn's Nurabot, Hexagon's AEON, plus component giants (Schaeffler backing Humanoid UK, Hexagon and reportedly Neura; LG investing in AgiBot and Dexmate while building its own CLOiD).
- **Data is the second product.** Nearly every serious player monetizes demonstration data: AgiBot World's million-trajectory dataset, Sunday's 2,000 paid glove-wearing "memory developers", 1X's security fleet feeding NEO, China's 40+ state-funded training centers. A hardware entrant without a data flywheel strategy competes on margins; with one, on compounding capability.
- **Certification favors wheels — for now.** Statically stable platforms certify under existing frameworks (ISO 13482, ISO 10218:2025, ANSI R15.08, UL 3300) while dynamically balancing robots await ISO 25785-1 (2026–27). The EU Machinery Regulation 2023/1230 (applies 20 Jan 2027, with AI-behaviour and cybersecurity requirements) is an opportunity for an EU vendor to lead on compliance-by-design.

**Bottom line for a German entrant:** the semi-humanoid category is real, growing, and — outside China — not yet locked up. The winning wedge combines (1) an industrial or care use case with demonstrable ROI, (2) compliance-by-design for the 2027 EU regime, (3) a data-collection strategy from day one, and (4) supply-chain positioning that exploits the new US restrictions on Chinese hardware rather than suffering them. Section 16 details the white spaces.

## 2 Scope, definition & methodology

### Definition

A **semi-humanoid robot** has a human-like upper body — head and/or torso with **two arms** — whose legs are replaced by another locomotion or mounting concept: a wheeled base (differential, omnidirectional/mecanum, swerve, two-wheel balancing or ball-balancing), a tracked base, a lift column, or a fixed pedestal. SoftBank's Pepper is the textbook example; AgiBot's A2-W, Galbot's G1, Rainbow's RB-Y1 and Sunday's Memo are the 2024–26 generation. Included statuses: shipping, announced, prototype, research and notable discontinued systems. Excluded: bipedal humanoids, single-arm mobile manipulators (documented in Annex A where influential — Toyota HSR, Hello Robot Stretch), arm-only cobots, armless telepresence robots, and dual-arm machines without an anthropomorphic upper body. Wheel-leg hybrids (Tencent The Five, GAC GoMate, Hexagon AEON, Moon Dynamics L1) sit in the annex as borderline.

### Method & data quality

Research was conducted in August 2026 through multi-lingual web research (English, Chinese, Japanese, Korean, German, French, Spanish, Russian): official vendor documentation and datasheets, trade press, financial media and filings, academic papers, community databases, teardown/forum material, and curated robot-description repositories. Every robot has a dossier (company profile, technical dossier, machine-readable spec sheet, source-attributed imagery — all images individually visually verified). **Every spec field carries a per-field source URL and one of three confidence ratings:** *vendor-claimed* (manufacturer statement), *third-party* (independently reported/verified), *estimated* (inferred). The companion xlsx preserves this per-field register (~3,400 source rows); the color coding in its matrix mirrors these ratings. Conflicting figures are retained with both values flagged rather than silently resolved. Where a vendor discloses nothing (a frequent Chinese and stealth-startup pattern), fields read "n/a (not disclosed)" — absence of a number is itself signal.

**Caveats.** This market moves at weeks-scale: 2026 alone brought the FCC action, Mobileye's \$900M Mentee acquisition, Walden's \$1.1B debut and several WAIC launches. Chinese shipment claims are frequently sell-in, not deployment (Interact Analysis: only ~10% of 2025 humanoid production reached real applications). Analyst forecasts diverge by >25× for 2035; we present ranges, not point estimates.

### Corpus at a glance

| Region | China | USA | Europe | Japan | Other | Korea | Canada |
|--------|-------|-----|--------|-------|-------|-------|--------|
| Robots | 60    | 41  | 36     | 31    | 21    | 15    | 3      |

| Status | shipping | discontinued | prototype | announced | research |
|--------|----------|--------------|-----------|-----------|----------|
| Robots | 82       | 38           | 38        | 31        | 18       |

Main-matrix corpus: 207 robots (plus 11 borderline/excluded in Annex A). "Other" combines Israel, India, Russia, Turkey, the Middle East, SE Asia and global/open-source projects. Canada is broken out in the xlsx but merged into the USA & Canada chapter here.

### 3 What the world calls this category

No single global term has settled; procurement, patent and market searches must cover all of the following. Chinese usage is the most developed — and China's Mobile Robot (AGV/AMR) Industry Alliance published a dedicated wheeled-humanoid Blue Book in May 2026 precisely because "the industry lacks unified standards and consensus."

| Language | Term                                                           | Literal meaning                         | Prevalence / notes                                                                                                                               |
|----------|----------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Chinese  | 轮式人形机器人                                                        | wheeled humanoid robot                  | <b>Dominant.</b> Used by UBTech, CASBOT, Topstar, Robot Era; industry analyses claim wheeled models are >80% of deployed "humanoids"             |
|          | 半人形机器人                                                         | semi-humanoid robot                     | Popularized by Pudu's D7 marketing; common in media                                                                                              |
|          | 轮式双臂机器人                                                        | wheeled dual-arm robot                  | Engineering/industrial framing (Fourier GRW, Rokae Helios)                                                                                       |
|          | 轮臂式机器人                                                         | wheel-arm robot                         | Emerging (Quectel, Zerith, KNEWBOTS)                                                                                                             |
|          | 复合机器人                                                          | composite robot                         | Legacy term for arm-on-AMR; adjacent category                                                                                                    |
|          | 超人形机器人                                                         | super-humanoid                          | New coinage for multi-arm forms (Midea MIRO U six-arm, iSageBot four-arm)                                                                        |
| Japanese | セミヒューマノイド                                                      | semi-humanoid                           | <b>Growing standard</b> in 2025–26 trade press (AGIRobots, Pudu coverage); replacing older 上体型                                                   |
|          | 車輪型ヒューマノイド                                                     | wheel-type humanoid                     | Standard trade-press term, esp. for Chinese imports                                                                                              |
|          | 双腕ロボット                                                         | dual-arm robot                          | Dominant industrial term (Yaskawa SDA, Kawasaki); mobility-agnostic                                                                              |
|          | 人型重機 / アバターロボット                                                | humanoid heavy machinery / avatar robot | Niche coinages: Jinki Ittai's railway robot; teleop category (ugo, Ory's 分身ロボット "alter-ego robot")                                               |
| Korean   | 이동형 양팔로봇                                                       | mobile dual-arm robot                   | Rainbow Robotics' official RB-Y1 category                                                                                                        |
|          | 휠 휴머노이드 / 세미 휴머노이드                                             | wheel humanoid / semi-humanoid          | Trade press; Seoul Economic Daily's March 2026 "wheeled semi-humanoids gain traction" wave coverage                                              |
| German   | semi-humanoider Roboter / humanoider Roboter auf Rädern        | —                                       | Media descriptions; also "mobiler Manipulator", "Pflege-/Assistenzroboter" (application terms), "kognitiver Roboter" (Neura/Oversonic marketing) |
|          | Roboter-Avatar                                                 | robot avatar                            | Telepresence subsegment (Devanthro Robody)                                                                                                       |
| French   | robot semi-humanoïde / humanoïde à roues                       | —                                       | Also "robot majordome" (butler), "robot d'accueil" (reception)                                                                                   |
| Spanish  | robot semihumanoide / humanoide con ruedas                     | —                                       | Also "manipulador móvil"                                                                                                                         |
| English  | wheeled humanoid / semi-humanoid / dual-arm mobile manipulator | —                                       | All three common; analysts add "mobile manipulation platform"; Morgan Stanley excludes wheeled torsos from "humanoid" entirely                   |

## 4 Market size & forecasts

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### The headline numbers — and why they mislead

Humanoid-robot forecasts span an extraordinary range: for 2035 alone, published market sizes run from ~\$9B (Precedence) through Goldman Sachs' \$38B TAM to ~\$250B (Future Market Insights) — a >25× spread driven by definitions, and long-horizon calls reach \$5T by 2050 (Morgan Stanley) and \$7T (Citi). Two structural facts matter more than any point estimate:

- **Semi-humanoids are under-counted.** Morgan Stanley's definition explicitly excludes "a humanoid torso mounted to a wheeled AMR and other bases that remain stable when powered down." The deployable end of the market is thus largely invisible in the most-quoted numbers — upside for anyone who counts properly.
- **Specialists are far more conservative than banks.** Interact Analysis (May 2026): >700k humanoid units/yr and ~\$15B revenue by 2035, commercial-viability inflection ~2032; 2025 production >20,000 units (10× 2024) but only ~10% deployed in real applications; China took >90% of production. Bank of America: 18k units 2025E → 1M/yr by 2030; BOM ~\$35k end-2025 falling to \$13–17k by 2030–35.

### What is actually known about the wheeled segment

- **China Blue Book (AGV/AMR Industry Alliance, May 2026):** China wheeled-humanoid shipments >5,000 units in 2025, market value >RMB 1.8B (~\$250M) — the only hard unit datapoint for the segment; implies wheeled ≈25–28% of Chinese humanoid volume [third-party; derivation estimated].
- **2025 humanoid actuals (all form factors):** Omdia counts 13,318 global shipments with Chinese vendors in the top six (AgiBot 5,100+, Unitree ~4,200–5,500, UBTech ~1,000); Xinhua/CSJ report ~18,000 China-only (+650% YoY). Counting rules differ; both are directionally consistent: China ≈90% of world volume.
- **2026 outlook:** GGII forecasts 62,500 China humanoid shipments; Morgan Stanley doubled its 2026 China forecast to 50,000 units in June 2026; TrendForce sees >700% YoY global growth. Chinese executives talk "100–200k level". Manufacturing deployments are described by China Securities Journal as "essentially all wheeled + dual-arm configurations, because ROI is decisive."
- **Commercial-report shares conflict irreconcilably** (wheeled = 40.6% GMI vs 85.3% Roots vs ~37% FMI) — use unit-level evidence, not percentage claims.

### Capital formation

Humanoid startups raised **\$6.1B across 139 deals in 2025** (PitchBook, +300% YoY) and **\$8.6B in the first seven months of 2026** — 1.8× all of 2025. China alone: ~RMB 39.8B across 325 deals in 2025 (+326%), and ≥RMB 100B invested in 2026 YTD — more than the previous five years combined — atop a ~RMB 1T (~\$138B) 20-year state fund and >RMB 70B of municipal funds. Semi-humanoid-relevant rounds include Galbot (~\$1.15B lifetime, \$3B valuation, backers from CATL to China's IC "Big Fund"), AgiBot (Tencent, BYD, LG; backdoor-listed via Shangwei), AI² Robotics (~\$735M at ~\$2.8–3B, July 2026), Atribot (unicorn, June 2026), Sunday (\$1.15B), Walden (\$300M seed at \$1.1B), Humanoid UK (\$1.35B), Dyna (~\$600M), Neura (Series C up to \$1.4B, Amazon/Nvidia participation, confirmed June 2026), and Mobileye's \$900M acquisition of Mentee. Government pull is Chinese policy: MIIT's 2023 guideline targets world-leading humanoid strength by 2027, with a reported 100,000-robot deployment goal.

## 5 Why wheels: the form-factor debate

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The engineering case for replacing legs is quantitative, and vendors increasingly make it openly:

- **Cost.** Galbot founder Wang He: a wheeled chassis is ~10× cheaper than legs and more stable; bipedal locomotion "unnecessary... would increase costs." Wheeled entry prices (\$4,290 Unitree R1-D; \$7,999 Weave Isaac; ~\$14k Astribot T1) sit an order of magnitude below bipedal industrial pilots.
- **Energy.** Legged electric locomotion suffers Joule-heating losses at low speed/high torque. Typical bipeds run 2–4 h (Optimus V2 ~2.3 kWh ≈2 h; Unitree H1 864 Wh <4 h static); wheeled semi-humanoids run 5–16 h and increasingly hot-swap (AgiBot A2-W: 2 kWh, 5 h, swappable; Galbot G1: 8 h; Dexmate Vega: 10–30 h claimed).
- **Safety physics & certification.** Rodney Brooks: falling legs carry high kinetic energy — his prediction is that "humanoid robots will get wheels for feet." A statically stable base stops rather than falls on power loss, and certifies under existing standards today (see §14). Korea's March-2026 wave (T-Robotics: the wheeled hybrid is our "final form... will never build legs") and the UK's Humanoid ("wheels first") made this an explicit strategy, not a compromise.
- **Engineering focus.** Budget not spent on balance goes into arms, hands and autonomy — the parts that earn revenue (1X's EVE-before-NEO data strategy; Dexterity's 60 kg-lift Mech).

**The counter-case** remains real: stairs, curbs and cable-strewn floors; legs as a positioning system (Agility); the "little teapot" instability of a static base reaching far under load (Melonee Wise); and household TAM arguments that pushed 1X from wheeled EVE to legged NEO. The pragmatic industry answer is increasingly modular lower bodies (Appttronik Apollo wheeled/legged variants; AgiBot X2-N leg-wheel switching) and wheel-leg hybrids (Hexagon AEON, GAC GoMate) — hedges that concede the near-term economy of wheels.

## 6 Regional landscape — China

**63 documented robots — the deepest, fastest, cheapest market.** Three strata: (1) **scale leaders** — AgiBot (5,100+ units 2025, 15,000 cumulative by June 2026, Fulin and Longcheer factory orders, backdoor-listed), Galbot (\$3B valuation; licensed robot pharmacies, ~170 unmanned kiosks, factory pilots at Mercedes-Benz China/CATL/ Zeekr), UBTech (Cruzr S2, exported to Japan via GA Robotics), Pudu and Keenon riding 100k/80k-unit service-robot install bases; (2) **technology specialists** — Astribot (10 m/s end-effector speed, \$14k T1), PaXini (~2,000 tactile sensors), Spirit AI (full force control), DexForce (Sim2Real), Robot Era (Haier's "HIVA" is its rebadged Q5); (3) **incumbent entrants** — Midea (six-armed MIRO U in its own washing-machine plant; owns KUKA), JAKA/Rokae/Elite/Topstar/Siasun from the cobot/industrial world, Fourier from rehab, plus ecosystem oddities from four-armed iSageBot to ByteDance's research ByteMini. A dedicated supplier layer (SEER's C1-D humanoid chassis, Realman's joint modules with claimed 50,000 h MTBF, Inspire hands) is commoditizing construction; data-collection platforms (ARX, AgileX, Galaxea, HEXFELLOW) arm the world's embodied-AI labs. Weaknesses: near-zero published safety certifications, sell-in vs deployment gaps, financial distress cases (CloudMinds), and now the US FCC wall.

| Robot / Company                                                       | Status   | Base                                                         | kg/arm | Price                                                                  | Deployment evidence                                                                                            |
|-----------------------------------------------------------------------|----------|--------------------------------------------------------------|--------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>AlphaBot 2</b> 爱宝<br>(Aibao)<br>AI² Robotics                       | shipping | wheeled mobile base with retractable waist-leg torso         | 10     | n/a                                                                    | Geely Jingneng Microelectronics semiconductor fab (03/2025-); Bloomage Biotech cleanrooms; automotive plants;  |
| <b>LIFT2 (ARX-LIFT2)</b><br>ARX-LIFT2<br>ARX Robotics                 | shipping | omnidirectional wheeled base with motorized lift column      | 5      | from CNY 249,800                                                       | wide adoption in Chinese research labs as ALOHA-style platform; no unit counts disclosed                       |
| <b>A2-W</b> 远征A2-W<br>AgiBot (Zhiyuan Robotics)                       | shipping | wheeled (4WD omni, zero turning radius, crab-walk)           | 5      | RMB 400k-800k (~USD 56k-112k) insider range, not officially published  | Fulin Precision ~100-unit order 08/2025 (tens of millions RMB); 4-robot demo 800+ totes/3h zero failures; batt |
| <b>Genie G2</b> 精灵G2<br>AgiBot (Zhiyuan Robotics)                     | shipping | wheeled (4-DoF omnidirectional chassis: omni, crab-walk, in- | 5      | not disclosed; predecessor Genie G1 at RMB 450k (~USD 62k)             | Longcheer ~1,000-unit framework order (hundreds of millions RMB) for tablet lines; Junpu Intelligent automotiv |
| <b>Cobot Magic</b><br>Cobot Magic<br>(Mobile ALOH<br>AgileX Robotics) | shipping | Tracer two-wheel differential AGV base, 685x570x155mm, 100kg | 1.5    | ~USD 48k class via distributors (India listing ₹3,999,999); config-dep | global research-lab adoption; AgileX Tracer base underpinned Stanford Mobile ALOHA                             |
| <b>S1</b> Astribot S1<br>Astribot (Stardust Intelligence)             | shipping | wheeled (3-DoF omnidirectional base) + flexing torso column; | 10     | research edition ~RMB 500k (~USD 69k); B2B quote; T1 sibling from RMB  | JD.com, CCTV, elder-care institutions, retail in six cities; reported thousand-unit Thundersoft order (unverif |
| <b>T1</b> Astribot T1<br>Astribot / Stardust Intelligence             | shipping | wheeled (compact base)                                       | 5      | from RMB 89,900 (~USD 13k); higher-spec variant ~RMB 99,000 reported   | first unit shipped 2026-06-01; mass production reported from 2026-07-09 (Wuxi); no named T1 customers public y |

| Robot / Company                                                                  | Status   | Base                                                            | kg/arm | Price                                                                  | Deployment evidence                                                                                            |
|----------------------------------------------------------------------------------|----------|-----------------------------------------------------------------|--------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>CASBOT W1 / W2</b><br>灵贰<br>CASBOT (Beijing Zhongke Huiling Robot Technology) | shipping | bipedal legs<br>(NOTE: discovery misclassified 02 as wheeled; t | n/a    | from RMB 328,800 on JD.com (02); 02 Lite dev platform RMB 99,000 promo | first JD orders within 48h of June 2025 listing; family robots working on Lenovo notebook assembly line; CASBO |
| <b>U05 (UU U05)</b> 优友<br>U05<br>Canbot                                          | shipping | three-wheel wheeled base, auto-docking charging                 | n/a    | HKD 320,000 (HK reseller, historical/out-of-stock)                     | hundreds of commercial deployments 2016-19 (banks, hospitals, airports, malls); National Day 70th anniversary, |
| <b>W1 Pro</b> 跨维W1<br>Pro<br>DexForce                                            | shipping | wheeled; omnidirectional (40 DoF config) or differential (34    | 10     | n/a                                                                    | vendor-listed partners: Huawei, BYD, Foxconn, Panasonic, Toyota, SOEs (aerospace/chemical/shipbuilding); coffe |
| <b>Mercury X1</b><br>Mercury X1 水星X1<br>Elephant Robotics                        | shipping | wheeled AGV base (torso = Mercury B1)                           | 1      | USD 15,999 (X1); Mercury B1 torso ~EUR 17,599 at EU reseller           | commercially listed at multiple international resellers since 2024; education/research users; volumes not disc |
| <b>R1 Pro</b> 星海图 R1<br>Pro<br>Galaxea AI                                        | shipping | wheeled omnidirectional; 3 self-developed steering-wheel mod    | 3.5    | USD 69,999 (US reseller Robotopian)                                    | research labs in CN + international; reference platform for Galaxea G0 VLA work; no industrial production depl |
| <b>G1</b> Galbot G1<br>Galbot (Galaxy General)                                   | shipping | wheeled omnidirectional base + single telescoping/folding le    | 5      | JD.com RMB 690-700k (~USD 97k); flash sale RMB 630k 02/2026; US resell | ~170 unmanned kiosk stores/pharmacies in 40+ cities; Haidian pharmacy license 2026-03-12; CATL/Mercedes-Benz C |
| <b>S1</b> Galbot S1 (银河通用)<br>Galbot / Galaxy General                            | shipping | four-wheel omnidirectional AMR base with mast                   | 30     | ~USD 55,000 listed (inquire; likely marketing anchor)                  | reported in regular use at CATL battery factory production lines (Jan 2026 onward)                             |
| <b>Juliet</b> 朱丽叶<br>Haohai Xingkong (HHS Robotics)                              | shipping | wheeled composite humanoid (concealed wheeled base)             | 10     | n/a                                                                    | Nanjing Road cabin (2 units, from 2026-02-13): 61,198 visitors, 2,368 orders, 3,082 items sold during Spring F |
| <b>e3 (e³-Mobile / e³-Desktop / e³-Mobile-π)</b> e³<br>HexFellow Robotics        | shipping | modular: TRIGGER-A3 3-omni-wheel holonomic (π variant) / MAV    | 2      | not disclosed (quote-based storefront)                                 | no named customers/unit counts; active GitHub (75 repos) and community                                         |
| <b>K1W</b> 节卡 K1W<br>JAKA Robotics                                               | shipping | wheeled omnidirectional base + lifting/pitching waist           | n/a    | n/a                                                                    | working in a Fortune-500 enterprise factory (sorting/handling/tending/inspection); 300+ embodied-family units  |
| <b>Kargo</b> JAKA Kargo (节卡)<br>JAKA Robotics                                    | shipping | omnidirectional wheeled AMR base with integrated cargo deck     | 5      | ~USD 65,000                                                            | CIIF 2025 launch; no named Kargo customers yet (company: 300+ embodied units 2024, >10k cobots installed)      |
| <b>XMAN-R1</b> XMAN-R1 (擎朗具身服务机器人)<br>Keenon Robotics                            | shipping | wheeled (2-DoF chassis, two wheel modules under kneeling-leg    | 3      | ~USD 100,000                                                           | Shangri-La Traders Hotel Shanghai Hongqiao commercial deployment (announced 2025-10-31) in multi-robot model w |

| Robot / Company                                                                                                  | Status   | Base                                                         | kg/arm | Price                                                                   | Deployment evidence                                                                                            |
|------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------|--------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>VersaBot VB1 / VB1-I / VB2</b><br>VersaBot (蓝芯科技)<br>Lanxin Robotics                                          | shipping | VB1/VB2: four-wheel omnidirectional AMR; VB1-I: dual-wheel d | 2      | ~USD 120,000 (third-party listing, unconfirmed)                         | Lanxin AMR/vision business deployed at Huawei, BYD, LG (vendor); no VersaBot-specific counts disclosed         |
| <b>MIRO U</b> 美罗U<br>Midea Group                                                                                 | shipping | wheeled(-legged) column chassis, 360° in-place rotation, six | n/a    | n/a                                                                     | Wuxi high-end washing-machine factory lines from end-2025 (own factory, self-reported 30% changeover gain); MI |
| <b>Hobbs / Hobbs W1</b> Hobbs W1 (松延动力)<br>Noetix Robotics                                                       | shipping | W1: wheeled omnidirectional platform (enclosed skirt base);  | n/a    | Hobbs bust ~USD 35,000; Hobbs W1 USD 73,500                             | 500+ JD.com pre-sale bookings in first days (Dec 2025); pre-launch deployments in museums, government halls, o |
| <b>Adam-U</b> Adam-U<br>PNDbotics                                                                                | shipping | fixed pedestal (lifting column + chassis, ~40 kg counterweig | n/a    | USD 45,000 (reseller in-stock)                                          | research labs / data-collection use; public demos (dance, cloth handling); volumes n/a                         |
| <b>TORA-ONE</b> TORA-ONE<br>PaXini Tech                                                                          | shipping | wheeled base, laser-SLAM navigation                          | 5      | n/a                                                                     | pilots claimed in logistics/industrial/commercial; no named customers; JD.com strategic investor; hands/sensor |
| <b>D7</b> PUDU D7 (半人形)<br>Pudu Robotics                                                                         | shipping | wheeled (omnidirectional chassis)                            | n/a    | not announced                                                           | no named customers or unit counts published; channel = 80k+ Pudu robots in 60+ countries                       |
| <b>FlashBot Arm</b> 闪电臂<br>Pudu Robotics                                                                         | shipping | wheeled delivery-robot base (FlashBot platform), 539x515mm f | n/a    | ~USD 34,000 (unofficial; base FlashBot ~\$28k; official pricing unpubli | no named FlashBot Arm customers public as of 08/2026; base FlashBot widely deployed in hotels                  |
| <b>Q5</b> 星动Q5<br>Robot Era                                                                                      | shipping | wheeled, with 3-DoF flexing/height-adjusting leg section     | 10     | n/a                                                                     | mall guidance deployments; Chengdu Panda Base; Chengdu debut 2025-06-19 with 100+ intent orders; company-wide  |
| <b>X1-PRO (with X1 predecessor and C1-D universal humanoid chassis)</b> 仙工 X1 PRO / C1-D 通用人形底盘<br>SEER Robotics | shipping | wheeled omnidirectional chassis with cargo buffer tray; sepa | 6      | n/a                                                                     | wheeled humanoids delivered to customer sites for trials since late 2024; logistics pilots; WRC 2025/CeMAT 202 |
| <b>Moz1</b> 小墨 (Xiaomo)<br>Spirit AI                                                                             | shipping | 4-wheel mecanum omnidirectional wheeled base                 | n/a    | ~USD 150,000 (aggregator estimate)                                      | CATL Zhongzhou battery-pack line: connector insertion for EOL/DCR testing, ~99% success, 3x efficiency (promot |
| <b>Cruzr S2</b> 克鲁泽S2<br>UBTech Robotics                                                                         | shipping | wheeled (full-size wheeled humanoid; drive type not disclose | n/a    | n/a                                                                     | WRC 2025 multi-robot warehouse demo; Japan commercial sales launch Oct 2025; no named end customers/unit count |
| <b>Wanda 2.0</b> 万达 2.0<br>UniX AI                                                                               | shipping | wheeled chassis (quiet, obstacle-avoiding; concealed drive w | n/a    | from RMB 88,000 (pre-sale base config, 2024-12-25)                      | CCTV-broadcast home demos (laundry, burgers, tofu, dishwashing) 08/2024; WRC 2024 booth; pre-sales + mass prod |

| Robot / Company                                                                | Status    | Base                                                         | kg/arm    | Price                                                       | Deployment evidence                                                                                             |
|--------------------------------------------------------------------------------|-----------|--------------------------------------------------------------|-----------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>R1-D</b> 宇树R1-D双臂机器人<br>Unitree Robotics                                    | shipping  | dual-arm upper body; fixed bench base (A5/A7) or wheeled bas | 2-4 (max) | from RMB 26,900 / USD 4,290 (R1-A5)                         | launch-quarter product, no named customers yet; company shipped >5,500 humanoids in 2025 (32.4% global unit)    |
| <b>Tian Yi 2.0</b> 天轶2.0 X-Humanoid (Beijing Humanoid Robot Innovation Center) | shipping  | wheeled AMR-style base with articulated torso pitch linkage  | n/a       | ~USD 60,000 (third-party listing)                           | Gold + silver medals, World Humanoid Robot Games Aug 2025 (27 autonomous 8mm precision insertions); JD.com ret  |
| <b>H1</b> Zerith H1<br>Zerith Robotics                                         | shipping  | universal-wheel compact chassis (corridor/elevator compatibl | n/a       | n/a                                                         | 20+ venues; tens of millions CNY orders, ~¥100M intention orders; 500-unit year-end target                      |
| <b>CASIVIBOT</b><br>CASIVIBOT (中科慧远)<br>CasiVision                             | announced | wheeled AMR base (enclosed, drive type n/a)                  | n/a       | ~USD 29,000 (configuration-dependent)                       | launch media coverage only; no named customers; obvious channel = CasiVision's installed AOI base               |
| <b>Centaur-G1</b><br>Centaur-G1<br>Elite Robots                                | announced | mecanum-wheel omnidirectional base, ±10mm chassis positionin | 5         | n/a                                                         | none announced; company has ~10,000 cobots in 50+ countries as conversion base                                  |
| <b>AICO 2</b> 爱可 AICO 2<br>Flexiv                                              | announced | dual-wheel differential-drive AMR base, laser SLAM           | 4         | n/a                                                         | no named AICO 2 deployments; MICO demoed ICRA 2026; company's Rizon arms widely deployed                        |
| <b>GRW</b> 傅利叶GRW 轮式双臂机器人<br>Fourier Intelligence                              | announced | wheeled (parallel mechanism, independent drive + steering pe | n/a       | n/a                                                         | WAIC 2026 booth demos only; no customers/pilots disclosed yet                                                   |
| <b>HIVA (Haiwa)</b> 海娃<br>Haier (platform: Robot Era)                          | announced | wheeled (Robot Era Q5 platform)                              | n/a       | n/a                                                         | AWE 2025 exhibition demos only                                                                                  |
| <b>KNEWBOTS wheel-arm robot</b><br>晓悟智能轮臂式机器人<br>KNEWBOTS                      | announced | wheeled AMR chassis                                          | 4.5       | n/a                                                         | no public deployments/unit counts; demo/pilot stage                                                             |
| <b>RealBot-S2 / RealBot-L2</b><br>RealBot-S2 / RealBot-L2<br>RealMan Robotics  | announced | S2: wheeled four-steer four-drive + folding leg; L2: wheeled | 5         | n/a                                                         | WAIC 2026 demos (fridge opening, cross-city teleop tea-making); company platforms in Yining Xinjiang restaurant |
| <b>R1</b> 蚂蚁灵波 R1<br>Robbyant (Ant Lingbo Technology, Ant Group)               | announced | wheeled pedestal (single-column, circular base)              | n/a       | from ~USD 55,000                                            | robot-chef demo (4 dishes) at Sep 2025 Bund Conference launch; reported pilots in Chinese museums, drugstores   |
| <b>Helios</b> 轮式双臂机器人Helios<br>Rokae Robotics                                  | announced | omnidirectional wheeled base (3 DoF planar)                  | 5         | n/a                                                         | none disclosed; trade-show demos (Automate 2026 Chicago)                                                        |
| <b>Onero H1</b> Onero H1<br>SwitchBot (OneRobotics)                            | announced | cylindrical wheeled base (vacuum-robot-style puck; drive lay | n/a       | USD 9,999 official store listing ('coming soon'); CES press | CES 2026 live demos only (laundry folding, dishes); pre-production 'lab' status; no customer deploym            |

| Robot / Company                                                                                                     | Status    | Base                                                         | kg/arm                            | Price                                                                   | Deployment evidence                                                                                            |
|---------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>TWH020 (Xiaotuo)</b><br>TWH020 小拓<br>Topstar Technology                                                          | announced | 4 steering-wheel (四舵轮)<br>omnidirectional chassis, 360°      | 10                                | n/a                                                                     | claimed first humanoid in Chinese injection-molding scenarios; deployments inside Topstar automation projects; |
| <b>SageMan (iSAGE STAR)</b> 赛智四臂人形机器人<br>SageBot/SageMa<br>iSAGEBOT /<br>Shanghai SAGE<br>Intelligent<br>Technology | announced | wheeled (AMR-derived platform)                               | 3-16<br>(configuration-dependent) | n/a                                                                     | no confirmed paid humanoid deployments; 100+ LOIs claimed for six-arm MAHAKBOT at CTIF Shanghai                |
| <b>AstroD. AD-01 (AstroDroid)</b><br>AstroDroid AD-01<br>(原力无限)<br>Elu.AI / INFIFORCE                               | prototype | wheeled AMR base with articulated pitch linkage to humanoid  | 3                                 | ~USD 100,000 (inquire)                                                  | APSARA Conference demo (2025-09-24) only; no named customers                                                   |
| <b>Futuring 2</b> 未来不远 二代<br>Futuring Robot                                                                         | prototype | wheeled base with slim column under dual-arm torso           | n/a                               | USD 5,000 (inquire)                                                     | claimed 300+ Chinese households, 30,000+ service hours (vendor-claimed, echoed by Chinese press)               |
| <b>Maker H01</b> Maker H01 (极佳科技)<br>GigaAI                                                                         | prototype | omnidirectional all-wheel-drive chassis                      | 5                                 | ~USD 160,000 (listed, unverified)                                       | none disclosed; unveiled 11/2025                                                                               |
| <b>ProWhite 2.0 (+ Zeus 1)</b> ProWhite Robot 2.0 (普罗宇宙)<br>PL-Universe                                             | prototype | four-wheel omnidirectional wheeled base                      | 7                                 | ProWhite: not disclosed (humanoid.guide \$45k entry conflicts with vend | none found; no named customers or pilots                                                                       |
| <b>Psi V1</b> Psi V1 (灵初智能)<br>PsiBot / Lingchu Intelligence                                                        | prototype | wheeled (rounded chassis, hub-motor style side wheels in ren | n/a                               | n/a                                                                     | small-scale warehouse sorting validation (vendor); mahjong/block demos; Meituan last-mile robot co-dev (separa |
| <b>Vinci</b> Vinci 轮式人形机器人<br>SVT Robot<br>(Hangzhou Spacetime Variable Technology)                                 | prototype | 4-wheel-drive / 4-wheel-steer omnidirectional wheeled base ( | n/a                               | n/a                                                                     | none public; office teleop demo videos only (late 2025)                                                        |
| <b>Songyi</b> 松羿<br>Siasun Robot & Automation                                                                       | prototype | wheeled (Siasun AMR-heritage mobile platform; drive type not | n/a                               | n/a                                                                     | WRC 2025 exhibition demos only; no customers disclosed                                                         |
| <b>I-Series (wheeled variant)</b><br>Simplexity Robotics                                                            | prototype | omnidirectional mecanum-wheel base with articulated waist li | n/a                               | USD 78,000                                                              | none; traction is financial (~\$289M in 5 rounds, unicorn, Tencent/Alibaba/HongShan)                           |
| <b>Quanta X2</b> 量子X2<br>X Square Robot                                                                             | prototype | wheeled (low sculpted chassis, concealed wheels)             | 6                                 | ~USD 80,000                                                             | demo videos only; no named customers found                                                                     |
| <b>Sean</b> Sean (ZeroTh)<br>Zeroth Robotics                                                                        | prototype | wheeled pedestal (rounded enclosed base, corner wheels)      | n/a                               | ~USD 65,000                                                             | none; absent from vendor's current product line (M1/W1/Jupiter) - likely internal/dev platform                 |

| Robot / Company                                                                                      | Status       | Base                                                         | kg/arm | Price                            | Deployment evidence                                                                                            |
|------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------|--------|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Zoomlion wheeled humanoid (unnamed, 2 models)</b> 中联重科轮式人形机器人<br>Zoomlion                         | prototype    | wheeled chassis with dual-arm humanoid torso (one variant ap | n/a    | n/a (internal use, not for sale) | dozens of humanoids working in Zoomlion Changsha Smart City factories (machining, logistics, pre-assembly, QC) |
| <b>ByteMini</b> ByteMini<br>ByteDance Seed                                                           | research     | wheeled mobile base with lift column                         | n/a    | n/a (internal research platform) | research demos only (table bussing, cloth manipulation, pick-and-place)                                        |
| <b>Ginger (XR-1 / Cloud Ginger 1.0 / 2.0)</b> Cloud Ginger<br>云端机器人小姜<br>CloudMinds / Dataa Robotics | discontinued | 3-wheel omnidirectional self-balancing chassis (skirted)     | n/a    | CNY 698,000 (JD.com listing)     | COVID field hospitals (Wuhan/Shanghai), telecom stores, schools, exhibitions 2019-2023; volumes undisclosed    |
| <b>Sanbot Max (King Kong)</b> Sanbot King<br>Kong<br>Qihan Technology                                | discontinued | four-wheel mecanum omnidirectional                           | n/a    | n/a                              | Sanbot line deployed at Shenzhen Bao'an airport and Gongbei Port of Entry (mainly Elf); international dis      |

## 7 Regional landscape — USA & Canada

**46 documented robots (incl. Canada) — the economics wave.** The 2025–26 US generation sells labor, not robots: Sunday's Memo (Coatue-backed, \$1.15B, glove-collected ~10M episodes, <\$10k target, beta by Thanksgiving 2026), Weave's \$7,999/\$449-mo Isaac, Robot.com's R-Noid and Eden's \$10/hour RaaS, Dyna's laundry robots at 99%+ 24-h success, Reflex's teleop-to-autonomy warehouse play (GXO; sub-\$50k target; Mexico factory), and Dexterity's 60 kg "superhumanoid" Mech (Kawasaki-built arms, Sanmina manufacturing). Walden Robotics crystallizes the moment: Russ Tedrake's TRI spin-out emerged July 2026 at a \$1.1B valuation with robots already in production shifts at a Toyota plant. Public-market reality check: Richtech (NASDAQ: RR) pairs a \$180k ADAM install base with \$5M revenue and governance flags. Canada contributes Sanctuary's hydraulic-hand Phoenix (now software-pivoted) and Kinova's discontinued MOVO. The deep research heritage — PR2, Baxter, Mobile ALOHA (\$32k BOM that spawned Sunday and the ALOHA-clone industry), Robonaut 2, TRI Punyo — remains the field's intellectual substrate. The FCC Covered-List action (§14) now shelters this market from foreign hardware — and pressures everyone else's US strategy.

| Robot / Company                                         | Status   | Base                                                         | kg/arm | Price                                                                   | Deployment evidence                                                                                            |
|---------------------------------------------------------|----------|--------------------------------------------------------------|--------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Aeo</b> aeo<br>Aeolus Robotics, Inc.                 | shipping | wheeled (drive type not disclosed; appears diff-drive with c | 3.6    | n/a (not disclosed); RaaS leasing/subscription only                     | since 2019: Medical Care Service Inc., Gakken Cocofump, HIMEDIC (eldercare); Tokyu Group, Globeship (property  |
| <b>Axol</b><br>Almond (Almond Robotics)                 | shipping | standalone / Axol Mount (fixed) / Axol Cart (wheeled)        | 6.5    | from USD 7,999; USD 11,999 full kit (cameras + Orin NX)                 | no named customers; batch sales to physical-AI teams; Product Hunt/press launch June 2026                      |
| <b>Alex</b><br>Boardwalk Robotics                       | shipping | pedestal (fixed) or third-party AMR mounting; no built-in lo | n/a    | n/a                                                                     | research-institution sales; aviation MRO pilot; carbon-fiber sanding demo                                      |
| <b>Simba (V1, Medium Payload Unit)</b><br>Deft Robotics | shipping | wheeled (holonomic) AMR base with lift column                | 4      | USD 34,900 (store 2026); earlier USD 40,000 + 30,000/yr maintenance; r  | 49 units sold (store counter); 2 paid enterprise contracts; Tier-1 US auto supplier case studies (compressor h |
| <b>Vega</b> Vega<br>Dexmate, Inc.                       | shipping | wheeled (omni-directional)                                   | 4.5    | USD 89,999 MSRP (\$999 preorder deposit); some third-party listings cit | in testing with unnamed publicly traded companies (manufacturing, logistics, retail); 6 Vega-U grant units to  |
| <b>Mech</b> Mech<br>Dexterity, Inc.                     | shipping | omnidirectional rover, 4 independently steerable wheels; spi | 29     | n/a                                                                     | truck-loading app launched 03/2025; Sanmina scaling deal for parcel/e-commerce/retail/air-cargo customers; uni |
| <b>DYNA-1</b><br>Dyna Robotics                          | shipping | stationary workcell (table/cart mounted); mobile base in dev | n/a    | n/a (not disclosed); RaaS monthly 'sell labor not hardware'             | hotels, commercial laundries, restaurants, gyms; 16+ h/day operation; 24h run 850-900+ napkins @ 99.4% success |
| <b>R2D3</b><br>Open Droids                              | shipping | wheeled AMR (Woosh chassis, diff-drive est.)                 | 5      | USD 55,000-60,000                                                       | rehab-center pilots (vendor-claimed); Baskin-Robbins/Subway pre-order talks; no named paying customers         |

| Robot / Company                                                                       | Status    | Base                                                         | kg/arm | Price                                                                   | Deployment evidence                                                                                             |
|---------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------|--------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>XLeRobot</b><br>Open-source project<br>(Vector Wang / Rice University RobotPi Lab) | shipping  | wheeled (3 omni wheels, holonomic; IKEA RASKOG cart chassis) | n/a    | from ~\$660 self-build; Wowrobo kit \$579 (CNY 3699)                    | ~5.4k GitHub stars, ~590 forks (2026-08); community builds, household-task demo videos                          |
| <b>Reflex</b> Reflex<br>Reflex Robotics                                               | shipping  | compact wheeled base, footprint <2x2 ft (<61x61 cm)          | n/a    | lease/RaaS only; reported target <USD 50,000/unit                       | GXO RaaS pilot since 2024-09-18 at Fortune 100 retailer omni-channel fulfillment site (tote transfers, picking) |
| <b>ADAM</b> ADAM<br>Richtech Robotics Inc.                                            | shipping  | stationary pedestal / counter-integrated workstation (no mob | n/a    | USD 180,000 list; RaaS USD 3,500/robot/month (25-unit/60-mo/\$5.25M dea | One Kitchen/Walmart ghost kitchens (Dawsonville GA 06/2024, Rockford IL 08/2024; 240-unit LOI; 20-restaurant f  |
| <b>TITAN</b><br>RoboForce                                                             | shipping  | interchangeable wheeled OR tracked mobile base (outdoor-rate | 40     | n/a                                                                     | Intersolar Feb 2025 debut; 2025 customer field pilots; Three Rivers Solar demonstration; 11,000+ pre-orders vi  |
| <b>R-noid</b><br>Robot.com (ex-Kiwibot)                                               | shipping  | holonomic wheeled base, 0.55 m footprint                     | 4      | RaaS subscription, rates not disclosed                                  | <40 units across ~12 customers (launch, 2026); Harbor Links Golf Course NY disclosed; kitchens, laundries,      |
| <b>Mobile AI</b> Trossen AI<br>Mobile Kit<br>Trossen Robotics                         | shipping  | wheeled (differential drive, SLATE UGV, 2x 350 W hub motors) | 1.5    | USD 33,695.95 list / 22,995 sale (no laptop); 37,845.95 list / 27,145   | successor to Stanford Mobile ALOHA; ALOHA-class kits widely used across academic labs, Google DeepMind and the  |
| <b>Operator OP1</b><br>Ultra (Ultra Robotics)                                         | shipping  | STATIONARY workcell on locking casters (no locomotion) - bor | n/a    | RaaS USD 2,500-3,000/month; 3-month trial                               | Highline Commerce (Brooklyn 3PL): ~30% of linear fulfillment workflow automated in live operation; 'wareh       |
| <b>Walden W0.5</b><br>Walden Robotics                                                 | shipping  | wheeled (legless; drive type not disclosed)                  | n/a    | n/a                                                                     | In production at a North American Toyota plant since 02/2026, 8-hour shifts; pilot-to-production <2 months;     |
| <b>Isaac 1</b> Isaac 1<br>Weave Robotics                                              | shipping  | wheeled base (3 DoF) with central column, passively stable   | n/a    | USD 7,999 up-front or USD 449/month; USD 250 refundable deposit         | no Isaac 1 field units yet; predecessor Isaac 0 (stationary folder) shipping since 02/2026: 2,000+ field hours  |
| <b>Dex</b> Dex<br>Richtech Robotics Inc.                                              | announced | wheeled AMR base (Richtech Titan delivery-series technology) | n/a    | n/a                                                                     | no customer announced as of 12/2025; demos: GTC 2025 DC baseball-cap assembly (booth 368), CES 2026             |
| <b>ROBIN</b><br>Roboligent Inc.                                                       | announced | wheeled AMR base                                             | n/a    | USD 85,000 (aggregator-listed, unverified)                              | no named customers; AFWERX STTR I + \$1.8M II fund Optimo rehab arm; RobotWorld 2025 Tesollo demo               |
| <b>Memo</b> Memo<br>Sunday Robotics<br>(Sunday Inc.)                                  | announced | wheeled platform with central telescoping column (no legs)   | n/a    | not announced; ~\$20k hand-built cost today; >=50% cost cut at scale    | no customer deployments yet; WIRED-observed live demos (espresso, dishwasher, glasses); 2,000+ paid glove data  |

| Robot / Company                                                   | Status       | Base                                                         | kg/arm | Price                                                                  | Deployment evidence                                                                                            |
|-------------------------------------------------------------------|--------------|--------------------------------------------------------------|--------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Syntro (V17)</b><br>WorkFar Inc.                               | announced    | 3-wheel omnidirectional skirted pedestal base                | n/a    | USD 9,999.99-59,999.99 by tier; USD 29,999 commonly cited; robot+opera | none independently verified; all claims from company PR; aggregator flags 'not verified'                       |
| <b>Beomni 1.0</b><br>Beyond Imagination, Inc.                     | prototype    | humanoid torso on wheeled mobile base or fixed workstation   | n/a    | n/a                                                                    | CES 2022 + Explorers Club teleop demos; 2022 SELF Labs LOI >=1,000 robots (no visible follow-through); no i    |
| <b>Eno</b><br>Genesis AI                                          | prototype    | wheeled base with foldable articulated tower torso           | n/a    | ~USD 90,000 (aggregator-reported, unconfirmed by vendor)               | LG CNS multi-phase pilot/scale partnership (LG US operations) announced at launch; no field deployments yet    |
| <b>KR1 / Kinisi 01</b><br>Kinisi Robotics                         | prototype    | wheeled (omnidirectional, zero-turn round base)              | n/a    | USD 75,000                                                             | no named customers; Automate Start-up Challenge winner May 2025; ~\$2M funding; acquired by Bear Robotics 2026 |
| <b>Eggie</b> Eggie<br>Tangible Robots                             | prototype    | wheeled (squat four-wheel base under column torso)           | n/a    | ~USD 32,000                                                            | demo videos only (laundry folding, spill cleaning, plant watering); no customers or pilots                     |
| <b>Tobi</b><br>TwoLabs                                            | prototype    | wheeled box base with articulated spine/waist column         | n/a    | USD 50,000 (target)                                                    | none; no pilots or customers found; 2-person YC Spring 2026 team                                               |
| <b>Mobile ALOHA</b><br>Stanford University (IRIS Lab)             | research     | wheeled (diff-drive, AgileX Tracer AGV)                      | 1.5    | ~USD 32,000 self-build BOM; Trossen Mobile AI kit ~USD 23-34k          | research only; replicated by many labs; DeepMind ALOHA 2 / Unleashed follow-ups; lineage to Sunday Robotics Me |
| <b>Punyo</b><br>Toyota Research Institute                         | research     | stationary upper-body rig (no mobile base published)         | n/a    | n/a                                                                    | lab demos (jugs, boxes, laundry baskets, handovers); peer-reviewed papers; Soft-Bubble design replicated by ot |
| <b>Monty</b><br>Anybots                                           | discontinued | two-wheel dynamically self-balancing                         | n/a    | n/a                                                                    | lab demos only (grasping glassware/dishes via teleop); balancing tech reused in QB telepresence robot          |
| <b>Astra (with Apollo wheeled/pedestal variants)</b><br>Apptronik | discontinued | Astra: fixed stand or third-party AMR. Apollo: modular lower | 10     | n/a                                                                    | Mercedes-Benz factory pilots since Mar 2024; Jabil plant pilot + contract manufacturing; GXO warehouse deploym |
| <b>HERB 2.0</b><br>CMU Personal Robotics Lab                      | discontinued | wheeled (Segway RMP 200, 2-wheel + casters), 655x620mm footp | n/a    | n/a                                                                    | single research platform; decade of manipulation-planning publications; Intel + NSF QoLT funded                |
| <b>Cody</b><br>Georgia Tech Healthcare Robotics Lab               | discontinued | wheeled omnidirectional (Segway RMP 50 Omni, mecanum), 750x6 | n/a    | n/a                                                                    | lab + patient studies only: bed-bath trials, robot-touch perception studies (2010-11)                          |
| <b>RoNA (Robotic Nursing Assistant)</b><br>Hstar Technologies     | discontinued | wheeled omnidirectional (mecanum)                            | 68     | n/a                                                                    | prototype demos and healthcare case studies only; TATRC/SBIR funded; no clinical deployment found              |

| Robot / Company                                                                     | Status       | Base                                                         | kg/arm | Price                                                                   | Deployment evidence                                                                                            |
|-------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------|--------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Domo</b><br>MIT CSAIL Humanoid Robotics Group                                    | discontinued | stationary upper torso (no mobility)                         | n/a    | n/a                                                                     | lab research only; 2007 PhD dissertation; lineage to Meka Robotics and Hello Robot                             |
| <b>M1 Mobile Manipulator</b><br>Meka Robotics                                       | discontinued | wheeled holonomic omnidirectional (Meka B1 base), 460x670mm  | n/a    | USD 320,000-340,000 depending on configuration                          | few units to research labs; Meka modules widely used (GT Cody, UT Austin Dreamer)                              |
| <b>Robonaut 2 (R2)</b><br>NASA JSC + General Motors                                 | discontinued | modular: ISS pedestal/stanchion; Centaur 2 4-wheel rover (Ea | 9      | n/a                                                                     | ISS 2011-2018 (task boards, switches; largely disabled 2015-18 by power fault); Centaur 2 desert trials 2010;  |
| <b>Guardian XT</b><br>Palladyne AI (formerly Sarcos Technology and Robotics)        | discontinued | platform-agnostic torso: mounts on tracked bases (Guardian X | n/a    | never commercialized; RaaS subscription planned (XO was ~USD 100k/yr)   | beta complete 2021, CES demos, Navy/USAF development work (incl. RE2 Sapien/HDMS); zero commercial deployments |
| <b>Baxter</b><br>Rethink Robotics                                                   | discontinued | stationary torso on optional caster pedestal (no powered mob | 2.2    | USD 22,000 base (2012); ~USD 25k with pedestal/options                  | thousands of Baxter+Sawyer shipped (vendor-claimed); SME packaging users; hundreds of university labs as defau |
| <b>uBot series (uBot-5/6/7)</b><br>UMass Amherst Laboratory for Perceptual Robotics | discontinued | two-wheel dynamically self-balancing (postural transitions t | n/a    | n/a                                                                     | lab research only: whole-body control, box flipping, postural transitions, first-responder teleop studies; bas |
| <b>BEAR (Battlefield Extraction-Assist Robot)</b><br>Vecna Technologies             | discontinued | tracked, articulated dual tracked 'legs' with dyna           | n/a    | n/a                                                                     | prototype demos: 227kg lift, dummy extraction, rubble traversal; soldier evaluations at Army Infantry Center ~ |
| <b>PR2</b><br>Willow Garage                                                         | discontinued | wheeled omnidirectional (4 powered caster modules), 668x668m | 1.8    | USD 400,000 (dual-arm); USD 285,000 single-arm SE; \$120k open-source d | 11 free beta units (2010) + sales; ~50 units total at research institutions worldwide; no commercial use       |

| Robot / Company                               | Status       | Base                                                         | kg/arm | Price                   | Deployment evidence                                                                                         |
|-----------------------------------------------|--------------|--------------------------------------------------------------|--------|-------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>MH3</b><br>Mirsee Robotics                 | announced    | wheeled AMR base (skirted, bumpered); wheels chosen over leg | 30     | USD 200,000             | none found; no customers, pilots or unit counts                                                             |
| <b>Phoenix (Generation 8)</b><br>Sanctuary AI | prototype    | wheeled (small mobile base; legs dropped as 'too frail       | n/a    | n/a (not for open sale) | Gen 5 retail pilot Jan 2023; Magna plant trials 2024-2025 (material handling, sub-assembly, <24h task learn |
| <b>MOVO</b><br>Kinova                         | discontinued | wheeled holonomic (4 Swedish wheels), 508x762mm footprint, A | 2.2    | n/a                     | university beta community: UWaterloo RoboHub, Georgia Tech CORE, UW-Madison, EU labs; est. dozens of units  |

## 8 Regional landscape — Europe

**38 documented robots — research superpower, fragmented commercially.** Active products span PAL Robotics' TIAGo Pro/++/ARI (the EU research standard), Pollen/Hugging Face's open-source Reachy 2 (~\$70k), igus's €47,999 plastics-based Iggy Rob, Oversonic's RoBee (STMicroelectronics deal, US launch — but certifications are CE-machinery grade, not robot-safety), Enchanted Tools' ball-balancing Mirokai in French hospitals, and Duatic's ETH-lineage Alpha (12 kg continuous dual-arm payload, IP66). Humanoid (UK) became Europe's newest unicorn (\$1.35B; Bosch and Schaeffler backing; wheeled Beta units Q4 2026); Franka's Mobile FR3 Duo gives European labs a domestic reference platform; Wandercraft's Calvin (bipedal, hence annexed) took Renault's money into real plants; and Munich's RobCo — fresh off a \$100M Series C — announced its bimanual, humanoid-form "Alfie" in April 2026 (officially shown only from the waist up — though its promo video briefly reveals a mobile base in simulation), moving Germany's best-funded modular-robot startup into the category via RaaS. Germany specifically: Neura's MiPA is the declared household/care entrant (€9,999, deliveries slipping; Series C up to \$1.4B confirmed June 2026), Devanthro sells care-avatar telepresence, and the research bench — DLR's Justin, KIT's ARMAR-III, TUM's GARMI (no spin-off exists — available IP/talent), FZI's HoLLiE — is unmatched but under-commercialized. Cautionary tales are equally instructive: Aldebaran (Pepper) liquidated in 2025 with its assets sold to a Chinese buyer; Care-O-bot's maker became an AMR vendor; Cybedroid sold 6 units and died. A telling blank on the origin map: the Nordics. Norway created the archetypal wheeled humanoid (Halodi/1X's EVE) but the company relocated to Palo Alto and pivoted to legs; Sweden's entry (Hexagon's AEON, Schaeffler-backed) is a wheel-leg hybrid sitting in this study's borderline annex; and Denmark's Odense cluster — Universal Robots' arms, MiR's bases — supplies the category's components without ever assembling a semi-humanoid of its own. Europe's gap is not capability but industrialization — precisely the entry a well-capitalized German robotics company can make ahead of the 2027 Machinery Regulation.

| Robot / Company                                                | Status   | Base                                                          | kg/arm | Price                                                                    | Deployment evidence                                                                                            |
|----------------------------------------------------------------|----------|---------------------------------------------------------------|--------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Mirokai</b> Mirokaï<br>(characters: Mir<br>Enchanted Tools) | shipping | ballbot (balances on a sphere driven by 3 omnidirectional wh) | n/a    | ~EUR 50,000 purchase + several hundred EUR/month subscription (2026);    | ISIR Sorbonne (first delivery Jul 2024); AP-HP incl. Broca & Annie Girardot pilots; Institut du Cancer de      |
| <b>Ameca</b> Ameca<br>Engineered Arts                          | shipping | fixed pedestal (non-walking; Gen 2.6 articulated non-walking) | n/a    | ~USD 130k-250k full body (quote-only); head configs from ~\$25k; custom  | National Robotarium Edinburgh, Computer History Museum, GSK trade shows, Sphere Las Vegas (EA robots); company |
| <b>QTrobot</b> QTrobot<br>LuxAI S.A.                           | shipping | stationary desk pedestal (no locomotion)                      | n/a    | Home: \$1,977 + \$149/mo support (2021); professional/research ~EUR 6,80 | therapy centers/schools/universities across EU + US + Asia (trials in LU/FR/BE/DE from 2018; Univ. Birmingham, |
| <b>Kime</b> Kime<br>Macco Robotics                             | shipping | fixed kiosk pedestal (~2 m2 enclosure); manually-moved cart   | 1      | n/a                                                                      | Seville bars (COVID contact-free beer service, 2020-21); Prio Portugal gas-station trial + planned rotation; S |
| <b>InMoov</b><br>Open-source project (Gael Langevin)           | shipping | none official (torso platform); community DIY wheeled bases   | n/a    | under USD 1,000 in parts (2013 estimate) + printing                      | thousands of community builds worldwide since 2012 (builder map, schools, universities, exhibitions)           |
| <b>RoBee</b> RoBee<br>Oversonic Robotics                       | shipping | wheeled (diff-drive; omnidirectional on request)              | 5      | n/a                                                                      | STMicroelectronics global supply deal (first unit in ST Malta fab, Dec 2025); several Italian factories; hospi |

| Robot / Company                                                                      | Status    | Base                                                         | kg/arm | Price                                                                  | Deployment evidence                                                                                            |
|--------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------|--------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>ARI</b> ARI<br>PAL Robotics                                                       | shipping  | wheeled (diff-drive, 2 DoF), skirt-covered base              | n/a    | not disclosed (quote-based)                                            | SPRING H2020: 7 units, Broca gerontology hospital Paris (AP-HP), 100+ patients, completed Oct 2024; Ayesa rece |
| <b>TIAGo++ / TIAGo OMNI++</b> TIAGo++<br>/ TIAGo OMNI++<br>PAL Robotics              | shipping  | wheeled; differential drive (TIAGo/++) or omnidirectional 4x | 3      | TIAGo++ from ~EUR 90,000 (2019, config-dependent); current pricing quo | 40 collaborative projects, 14 EU countries, 27 countries worldwide (vendor); RoboCup@Home 2019 winner platform |
| <b>TIAGo Pro</b> TIAGo Pro<br>PAL Robotics                                           | shipping  | wheeled omnidirectional (4x mecanum wheels, 4 DoF)           | 3      | not disclosed (quote-based); ~USD 85-95k reported in market chatter, u | CO-HAND EU project: TRL 7 validation in textile manufacturing (2026); ROSALYA food-industry project; euROBIN r |
| <b>Reachy 2</b> Reachy 2<br>Pollen Robotics (Hugging Face)                           | shipping  | wheeled omnidirectional (3 omniwheels); stationary mount opt | 3      | ~USD 70,000 (dual-arm + mobile base)                                   | Cornell University, Carnegie Mellon University research labs; internal Hugging Face/LeRobot use; unit count un |
| <b>Steve</b><br>Sulu.be (Jan De Coster studio)                                       | shipping  | wheeled cart base (large front drive wheels + casters); huma | n/a    | USD 60,000                                                             | character host in Belgian movie theaters (Brightfish), corporate receptions, festivals, brand activations; uni |
| <b>iggyRob</b> iggyRob (Iggy Rob)<br>igus GmbH                                       | shipping  | wheeled AMR (ReBeLMove Pro, differential drive, three-point  | 2      | EUR 47,999 (~USD 54,500); iggyRob Home variant price on request        | no named external customers; igus internal use for injection-molding machine tending planned; live catalogue a |
| <b>MiPA</b> MiPA (My intelligent Per<br>NEURA Robotics GmbH                          | announced | wheeled (actuated wheels, geometry not disclosed) + linear-a | 3      | EUR 9,999 introductory (MiPA Home, private households only); EUR 100 r | none verified; trade-fair demos only (Automatica 2025 beverage service, IFA 2025 tidying, CES 2026 dishwasher) |
| <b>Alfie</b><br>RobCo                                                                | announced | mobile base shown in simulation (promo video 0:14): broad fl | n/a    | RaaS, terms undisclosed                                                | none yet (Hannover Messe preview)                                                                              |
| <b>pib.Pro</b> pib (printable intellige<br>isento GmbH / isento robotics (pib.rocks) | announced | two-wheeled self-balancing (hoverboard-style) mobile base un | n/a    | USD 5,999                                                              | DIY pib in German schools via pib.Education; no named pib.Pro customers yet                                    |
| <b>Clone Torso</b><br>Clone Torso<br>Clone Robotics                                  | prototype | none — stationary mounted upper body/torso (full android Pro | n/a    | not published for Torso; manufacturing cost <\$20k/unit (vendor clai   | none beyond vendor videos (viral Torso/Protoclone clips); no independent evaluations or customer deployments f |
| <b>Robody</b> Robody<br>Devanthro GmbH (The Robody Company)                          | prototype | wheeled mobile base (drive type not disclosed)               | n/a    | n/a — care-service model (Robody Cares), pricing undisclosed           | Private-home deployments since early 2024 (3+ hw generations); Teleskoop BMBF project with FZI/Charité: househ |
| <b>Alpha</b> Alpha<br>Duatic AG                                                      | prototype | wheeled (mecanum omnidirectional)                            | 6      | n/a (not disclosed); USD 150k third-party estimate                     | 20 kg bag autonomous lift demo video; DynaArm demos on ANYmal and Festo BionicMobileAssistant; no named custom |
| <b>Eden I</b><br>Eden Robotics, Inc.                                                 | prototype | floor-standing rounded base + lift column (seen in site vide | n/a    | RaaS from USD 10/hour, USD 0 setup; no purchase option                 | none public as of Aug 2026                                                                                     |

| Robot / Company                                                                                                               | Status    | Base                                                                     | kg/arm | Price                                                                           | Deployment evidence                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Mobile FR3 Duo</b><br>Franka Robotics                                                                                      | prototype | wheeled<br>(omnidirectional<br>TMR base,<br>800x580x294 mm)              | 3      | not published (early<br>access)                                                 | Automatica 2025 showcase; no<br>commercial deployments disclosed                                                         |
| <b>HMND 01<br/>(wheeled) / Alpha<br/>Wheeled</b> HMND<br>01 Alpha Wheeled<br>Humanoid<br>(Humanoid Ltd /<br>SKL Robotics Ltd) | prototype | wheeled<br>(omnidirectional<br>pedestal base)                            | n/a    | n/a (not disclosed); RaaS<br>subscription incl. fleet<br>software, maintenanc   | PoCs: Ford Cologne (97% pick<br>reliability, 83 picks/h, third-party);<br>SAP+Martur Fompak (autonomous,<br>SAP EWM-dri  |
| <b>Alfie</b> Alfie<br>Prosper Robotics                                                                                        | prototype | wheeled (indoor,<br>single-floor)                                        | n/a    | target ~USD 5,000-15,000<br>hardware + ~USD<br>500/month subscription<br>(maint | prototypes tested in Prosper<br>employees' homes; no customers or<br>unit counts; launch slipped from<br>~2025           |
| <b>ALPHA<br/>(Automated<br/>Litter Processing<br/>Humanoid<br/>Assistant)</b><br>TeknTrash                                    | prototype | wheeled AMR base<br>with tall lift column<br>and integrated collec       | n/a    | USD 55,000                                                                      | pilot deployments with waste-<br>management operators claimed; no<br>named sites or counts                               |
| <b>Northstar</b><br>Northstar<br>UMA                                                                                          | prototype | wheeled (V0 initial<br>platform; bipedal<br>only on later<br>roadmap     | n/a    | ~USD 70,000 (discovery<br>sweep, unconfirmed)                                   | none yet; ~50 potential customers<br>evaluating (vendor-claimed); tethered<br>prototype at July 2026 unveiling           |
| <b>Rollin' Justin</b><br>Rollin' Justin (Agi<br>DLR Institute of<br>Robotics and<br>Mechatronics                              | research  | wheeled, 4<br>independently<br>steered spring-<br>suspended wheels<br>on | 14     | n/a (not for sale)                                                              | METERON SUPVIS-Justin ISS<br>teleop 2017 (Nespoli); Surface<br>Avatar ISS campaign 2023-2025<br>(Rubio, Dyson/Epps, Kim  |
| <b>HoLLiE (HoLLiE<br/>C)</b> HoLLiE —<br>House of Living<br>FZI Research<br>Center for<br>Information<br>Technology           | research  | Clearpath<br>Ridgeback<br>omnidirectional<br>(mecanum)<br>commercial pla | n/a    | n/a (not for sale)                                                              | HoLLiECares (BMBF): two field-test<br>phases in Städtisches Klinikum<br>Karlsruhe and Knappschaftsklinikum<br>Saar (~20  |
| <b>R1</b> R1 — your<br>personal human<br>Istituto Italiano di<br>Tecnologia (IIT)                                             | research  | two driving wheels<br>(diff-drive) + 3<br>pivoting 75 mm<br>castors      | 1      | target 'cost of a small<br>family car' (~EUR 20-30k<br>class, esti              | rehabilitation-assistant trials with<br>Fondazione Don Gnocchi (assistive-<br>rehab joint lab); HR1 healthcare<br>assist |
| <b>ARMAR-7</b><br>ARMAR-7<br>KIT H2T (High<br>Performance<br>Humanoid<br>Technologies Lab)                                    | research  | holonomic wheeled<br>platform, 3 omni-<br>wheels                         | n/a    | n/a (not for sale)                                                              | Automatica 2025 + Hannover Messe<br>2026 live demos (dishwasher<br>unloading, laundry sorting, learning<br>from demonst  |
| <b>GARMi</b> GARMi<br>TUM MIRMI<br>(Geriatrics)                                                                               | research  | wheeled mobile<br>base (stable, low<br>CoG) + extendable<br>lifting c    | n/a    | n/a (not for sale)                                                              | model apartment trials at<br>Geriatrics centre Garmisch-<br>Partenkirchen with Caritas/LongLeif;<br>telemedicine and r   |

| Robot / Company                                                                               | Status       | Base                                                         | kg/arm | Price                                                            | Deployment evidence                                                                                            |
|-----------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------|--------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Alter-Ego (Alter-Ego X)</b> ALTER-EGO<br>Univ. Pisa Centro Piaggio + IIT SoftBots          | research     | two-wheel self-balancing (inverted pendulum)                 | n/a    | n/a (not for sale)                                               | pilot clinical avatar study at IRCCS Maugeri Milan (Fit4MedRob/PNRR): patient welcome, pain assessment, accomp |
| <b>EVE</b> EVE<br>1X Technologies (ex-Halodi Robotics)                                        | discontinued | two-wheel self-balancing wheeled base, multi-terrain wheels  | n/a    | RaaS: ~NOK 500,000 (~USD 50k)/unit/year under 2022 ADT agreement | Everon/ADT EvoGuard night guarding, 140 units/yr lease (2022), est. 150-250 units cumulative; pilots: Altopack |
| <b>Leenby</b> Leenby<br>Cybedroid                                                             | discontinued | wheeled base under skirt (drive type not disclosed)          | n/a    | n/a                                                              | ~10 units total by 2019: 6 sold (companies, nursing homes, research), 3 event rentals, 1 six-month Crédit Agri |
| <b>AILA</b> AILA<br>(Artificial Intelli DFKI Robotics Innovation Center)                      | discontinued | six-wheeled omnidirectional base, each wheel a 2-DoF (drive+ | n/a    | n/a (never for sale)                                             | research projects only: SemProM, Robofoot (2010-2013), BesMan (2012-2016, ISS mockup/SpaceBot Cup), HySociaTea |
| <b>Care-O-bot 4</b><br>Care-O-bot 4 ("Paul Mojin Robotics (now 4am Robotics) / Fraunhofer IPA | discontinued | wheeled omnidirectional                                      | n/a    | USD 66,000-232,000 by configuration (historical)                 | 'Paul' greeter at Saturn stores from Oct 2016 (Ingolstadt first); museum/airport/care trials; small            |
| <b>AMIGO</b> AMIGO<br>(Autonomous Mate f Tech United, TU Eindhoven                            | discontinued | four omni wheels (soccer-robot drive technology); base house | 1.5    | n/a (self-built)                                                 | RoboCup@Home: Dutch Open champion 2012, world runner-up 2014 (Brazil); RoboEarth + Bobbie Robotics demonstrato |
| <b>Cosero</b> Cosero<br>(Cognitive Servic University of Bonn — NimbRo                         | discontinued | omnidirectional: 4 individually steerable wheel-pair axes, 8 | n/a    | n/a (self-built)                                                 | RoboCup@Home world champion 2011-2013 + German Opens 2011-2014; demos: omelet cooking, fridge access, table ca |

## 9 Regional landscape — Japan

**33 documented robots — the longest institutional memory, now re-mobilizing.** Japan invented the category's ancestors (wakamaru 2005, enon, SmartPal, Twendy-One, ROBEAR) and its icon: Pepper (~27,000 produced — still the largest semi-humanoid fleet ever; B2B support continues in Japan, with a "Pepper+" rental relaunch in 2026 even as Aldebaran died in Europe). Today's activity clusters in: **teleoperation-first services** (ugo Pro in building security; Ory's OriHime-D café employing 70+ disabled pilots — a social-innovation model unique to Japan; Jinki Ittai's "humanoid heavy machinery" in revenue service on JR West railways); **industrial dual-arm incumbents** (Kawada NEXTAGE since 2009; Yaskawa's MOTOMAN NEXT AI line — and note Yaskawa wholly owns Tokyo Robotics, whose Torobo2 builds Waseda's AIREC care robot: one keiretsu quietly spans Japan's flagship care program and its industrial dual-arm line); **hospital/care pilots** (Kawasaki Nyokkey at Fujita Health University; Foxconn/Kawasaki Nurabot launching FY2026); and **Toyota's ELEY** — QDD actuators in every joint, a scapular axis, and a production-site learning agenda: the highest-credibility incumbent program worldwide (§12). Tracked outlier: Mitsubishi Electric's Diaroid. Japanese firms also increasingly distribute Chinese wheeled humanoids (Cruzr S2 via GA Robotics), importing the competition.

| Robot / Company                                                                                                                                                                        | Status   | Base                                                          | kg/arm | Price                                                                  | Deployment evidence                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------|--------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Zero-shiki Jinki ver.2.0 / Multifunctional Railway Heavy Machine</b> 零式人機 ver.2.0 / 多機能鉄道重機<br>Jinki Ittai (developer/licensor) + Nippon Signal (manufacturer) + JR West (customer) | shipping | humanoid torso on telescoping crane boom of road-rail vehicle | n/a    | n/a                                                                    | revenue service on JR West operating lines since 07/2024 (painting, tree cutting); operated by Nishi-Nippon EI |
| <b>NEXTAGE / NEXTAGE Fillie</b> ネクステージ / ネクステージ フィリー<br>Kawada Robotics                                                                                                                | shipping | fixed pedestal (manual repositioning between cells; no mobil  | 2      | Fillie from ¥6,000,000 (pedestal, hand cameras optional extra)         | 100+ companies since 2009 (third-party DB); electronics/auto/cosmetics/food; GLORY Saitama cell lines (est.);  |
| <b>OriHime-D</b> オリヒメ D (分身ロボット)<br>OryLab Inc. (Ory Laboratory)                                                                                                                       | shipping | wheeled base (forward/backward/rotate, teleoperated)          | n/a    | n/a (service/employment model, not sold retail)                        | Avatar Robot Cafe DAWN ver.B permanent in Nihonbashi since 2021-06-21; 70+ disabled pilots employed (02/2024); |
| <b>Foodly</b> フードリー<br>RT Corporation                                                                                                                                                  | shipping | pedestal on casters (manual repositioning, no self-drive)     | n/a    | n/a (made-to-order, customized); monthly rental available              | Hirai (Kumamoto) bento line; Ichibiki (Aichi) retort side-dish line; Will of Factory staffing partnership; MET |
| <b>Sciurus17</b> シューラス17<br>RT Corporation                                                                                                                                             | shipping | bench-mount torso; optional S17-LIFT 700mm lift column docki  | 0.5    | JPY 3,300,000 base; OP1 head JPY 55,000; Lift and Kachaka base separat | standard lab platform at Japanese universities/companies since 2018; unit counts not disclosed                 |
| <b>SEED-Noïd (SEED-R7 series)</b> シードノイド (SEED-R7シリーズ)<br>THK / SEED Solutions                                                                                                         | shipping | modular: SEED-Mover omnidirectional cart (4 omni wheels) + S  | n/a    | n/a (quotation-based; units sold individually)                         | NTT Communications ExTorch program adoption (2021-05); WRS-era research platform; academic use; no public unit |

| Robot / Company                                                                               | Status    | Base                                                         | kg/arm | Price                                                        | Deployment evidence                                                                                            |
|-----------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------|--------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Torobo (Torobo2)</b><br>トロボ<br>Tokyo Robotics Inc.                                         | shipping  | wheeled omnidirectional cart (4-axis)                        | 7      | n/a (quotation)                                              | shipping to research labs; SusHi Tech Tokyo 2026 exhibit; no public unit counts                                |
| <b>Robovie-R3</b> ロボビー・アール3<br>(Robovie-R Ver Vstone (ATR lineage))                           | shipping  | wheeled differential drive (2x40W); optional omnidirectional | n/a    | JPY ~3,800,000                                               | widely used in Japanese HRI research (ATR field trials in malls, schools, museums); care-facility and retail e |
| <b>ugo Pro</b> ugo Pro (ユーゴー プロ)<br>ugo, Inc. (ex-Mira Robotics)                              | shipping  | wheeled (drive type n/a; diff-drive est.) + torso lifter     | n/a    | n/a (RaaS via partners, 'negotiable'); armless sibling ugo E | Teikyo Univ Itabashi (05/2024), Yanmar FLYING-Y Bldg Osaka (11/2024), Mainichi Newspaper Bldg Osaka (04/2025), |
| <b>AGIRobots Worker</b><br>AGIRobots Worker (車輪型セミヒ AGIRobots Inc.                            | announced | wheeled — swerve-type steered-wheel modules, omnidirectional | 4      | n/a (not disclosed)                                          | none; basic development complete; Robot Technology Japan 2026 live demo planned                                |
| <b>Modular</b><br>O-ID                                                                        | announced | wheeled base with wheel hub motors (flat-floor); humanoid up | n/a    | USD 70,000                                                   | none found                                                                                                     |
| <b>MOTOMAN NEXT-NHC10DE</b><br>MOTOMAN NEXT-NHC10DE<br>Yaskawa Electric                       | announced | stationary dual-arm (pedestal/cell mount; no mobile base sho | 10     | n/a                                                          | iREX 2025 picking/packing demos only; no customers announced. Lineage traction: MOTOMAN-SDA 15-axis dual-arms  |
| <b>Nurabot</b> Nurabot (護理機器人)<br>Foxconn (Hon Hai) + Kawasaki Heavy Industries, AI by NVIDIA | prototype | wheeled AMR (Nyokkey-derived) + dual-arm torso + cargo compa | n/a    | n/a                                                          | Taichung Veterans General Hospital field trials since 04/2025 (delivery, guidance, night patrol); claimed 20-3 |
| <b>Nyro</b><br>Individual maker (Ryota Kobayashi)                                             | prototype | none (upper body on camera tripod; no mobile base yet)       | 0.5    | n/a                                                          | none; project page 436 views / 5 likes (niche hobbyist visibility)                                             |
| <b>Nyokkey</b> ニョッキー<br>Kawasaki Heavy Industries / Kawasaki Robotics                         | prototype | wheeled AMR base + humanoid torso                            | 6      | not for sale (pilot placements only)                         | Fujita Health Univ Okazaki Medical Center: specimen transport + auto-loading into clinical analyzers with FORR |
| <b>DiaroiD</b> ディアロイド<br>Mitsubishi Electric                                                  | prototype | tracked — CuboRex CuGo MEGA electric crawlers (independent b | 20     | n/a (not commercialized)                                     | no commercial deployments; verified tasks: plug insertion ~4min, M6 bolt fastening, zipper-bag opening ~2min;  |
| <b>Honda Avatar Robot</b> Honda アバターロボット<br>Honda Motor Co., Ltd.                             | research  | wheeled mobile base under dual-arm upper body                | n/a    | n/a (research program)                                       | none; press demos (pull-tab can, ball force control, tool use) 2022; FY2023 technology verification            |
| <b>ibuki</b> イブキ<br>Osaka University<br>Ishiguro Lab / JST ERATO                              | research  | wheeled mobility unit: 2 active wheels + eccentric wheels (h | n/a    | one-off research platform, never sold                        | lab and public demonstrations (hand-holding walk demos, telepresence studies); no deployments                  |
| <b>Panda dual-arm mobile manipulator</b><br>PRIN Lab (NII)                                    | research  | omnidirectional (TriOrb ball-drive base)                     | 3      | n/a (self-integrated)                                        | lab use; public demo video                                                                                     |

| Robot / Company                                                                                                           | Status       | Base                                                         | kg/arm | Price                                                                   | Deployment evidence                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------|--------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>ELEY</b> エレイ<br>(Embodied LEarning r<br>Toyota                                                                         | research     | wheeled (rounded mobile base per official render)            | n/a    | n/a (research)                                                          | WRS 2025 Aichi exhibit; production-site pilots planned; HSR-derived tech already fielded (Potaro hospital tran |
| <b>AIREC (Dry-AIREC ver. 1.0)</b> アイレック<br>(AI-driven Robot f<br>Waseda University<br>Sugano Lab / JST<br>Moonshot Goal 3 | research     | wheeled round mobile base with articulated single-column low | n/a    | target >= JPY 10M (~USD 67k) at ~2030 introduction                      | care-facility trials since 2024; demos: patient rolling, supine-to-sit transfer, sock dressing, scrambled eggs |
| <b>TWENDY-ONE</b> トウエンディワン<br>Waseda University<br>Sugano Laboratory                                                      | research     | omnidirectional wheeled cart                                 | n/a    | n/a (never sold)                                                        | none commercial; landmark demos 2007-2015: bed-to-wheelchair transfer assist, breakfast preparation, handovers |
| <b>enon</b> エノン<br>Fujitsu Frontech /<br>Fujitsu Laboratories                                                             | discontinued | wheeled (two drive wheels, 2 DoF per wheel)                  | 0.5    | individual quotation; no list price                                     | pilot deployments: shopping-center guidance, showrooms, Kyotaro Nishimura Museum guide (2007); unit count neve |
| <b>G1</b> GITAI G1<br>GITAI                                                                                               | discontinued | humanoid torso, fixed/rail mount (microgravity-oriented, no  | n/a    | n/a (never sold)                                                        | JAXA Kibo mock-up test 2018-12: 72% (13/18) astronaut tasks completed; JAXA joint research 2019-03; no flight  |
| <b>MELTANT-beta</b><br>MELTANT-β (メル<br>タント・ベータ)<br>MELTIN MMI                                                            | discontinued | pedestal/frame-mounted upper body (no self-locomotion)       | n/a    | never sold                                                              | field trials in construction/plant settings; JAXA ISS crew-task avatar tests; AVATAR X program (ANA/JAXA); G20 |
| <b>wakamaru</b> ワカマル<br>Mitsubishi Heavy<br>Industries                                                                    | discontinued | wheeled (two drive wheels + casters), 450 mm base diameter   | 0      | JPY 1,575,000 incl. tax (2005 retail); rental JPY 20,000-120,000/day (  | 100-unit household test sale (Tokyo, 2005); ~30 event deployments incl. Expo 2005 Aichi; corporate reception r |
| <b>ROBEAR</b> ロベア<br>RIKEN / Sumitomo<br>Riko                                                                             | discontinued | wheeled base with extendable stabilizing outrigger legs      | n/a    | never sold                                                              | lab demonstrations only (patient lifting, standing assistance); no field deployment                            |
| <b>WorkSense W-01</b><br>ワークセンス W-01<br>Seiko Epson                                                                       | discontinued | passive wheeled stand (manually repositioned; no autonomous  | n/a    | quotation-based, never published                                        | at least one global manufacturing customer piloting in Europe (2018); no install-base figures published        |
| <b>Pepper</b> ペッパー<br>SoftBank Robotics<br>(orig. Aldebaran /<br>SoftBank Robotics<br>Europe)                             | discontinued | wheeled (3 omnidirectional wheels)                           | n/a    | 2015: ¥198,000 + 36-mo subscription ≈ ¥1.08M 3-yr; Biz ~\$440-550/mo; 2 | ~27,000 produced / ~17,000 sold, 70 countries; Hamazushi all branches, banks, Pepper PARLOR cafe, Prague &     |
| <b>Model-T</b> モデルT<br>Telexistence Inc.                                                                                  | discontinued | wheeled base + vertical axis under dual-arm torso            | n/a    | n/a (service model, never sold)                                         | FamilyMart Toshima Ecomusee Town trial (summer 2020); Lawson Tokyo Port City Takeshiba (opened 2020-09-14); su |
| <b>Robina</b> ロビーナ<br>(TPR-Robina)<br>Toyota Motor<br>Corporation (Partner<br>Robot division)                             | discontinued | wheeled under skirt: 2 drive wheels + 4 maneuvering wheels;  | n/a    | never sold (2 units built)                                              | continuous public tour-guide service at Toyota Kaikan Exhibition Hall 2007-2009 (+twin thereafter) — among the |

## 10 Regional landscape — South Korea

**16 documented robots — chaebol-scale bets plus an explicit 2026 wheeled wave.** Rainbow Robotics' RB-Y1 (\$80k research/\$120k commercial) is the flagship: Samsung took 35% control for \$181.6M, consolidated it, and stood up a Future Robotics Office; Coupang logistics pilots are confirmed. ROBOTIS ships the fully open-source AI Worker (swerve drive, imitation-learning workflow); XYZ's DEUX (\$29,900 fixed / \$39,900 mobile, 3-finger hands + Glove X data pipeline) targets retail restocking; LG's CLOiD (CES 2026, "Zero Labor Home", pilot production July 2026) attacks the home — while LG hedges via investments in AgiBot and Dexmate. The March-2026 "wheeled semi-humanoid wave" (Seoul Economic Daily) added listed incumbents T-Robotics ("will never build legs") and I.L.'s ILBOT (mass production since Feb 2026), plus Neubility's announced Billy and Neuromeka's cart-based EIR. Hyundai's DAL-e and legacy research (KIST, KAIST M-Hubo, NAVER AMBIDEX) round out a market whose defining feature is vertical integration into memory-chip-grade manufacturing supply chains.

| Robot / Company                                      | Status    | Base                                                         | kg/arm | Price                                                                  | Deployment evidence                                                                                            |
|------------------------------------------------------|-----------|--------------------------------------------------------------|--------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>AI Worker (FFW-SG2 / FFW-BG2)</b> AI워커<br>ROBOTIS | shipping  | wheeled (swerve drive, 3 independent steer+drive modules) —  | 3.0    | FFW-BG2 ~USD 40,000 incl. VAT (EU reseller); quotes via ROBOTIS Americ | Reuters May 2026: trained on skilled workers' motions for industrial tasks; K-Humanoid Alliance member; u      |
| <b>RB-Y1</b> 이동형 양팔로봇 RB-Y1<br>Rainbow Robotics      | shipping  | wheeled (2-wheel drive; optional Mecanum omnidirectional kit | 3      | USD 80,000 (research) / USD 120,000 (commercial), VAT excl.            | 130+ cumulative units (2025); MIT/UC Berkeley/UW/Georgia Tech; Coupang fulfillment-center pilot CONFIRMED (fir |
| <b>ALICE M1</b> 앨리스 M1<br>A Robot / AeiROBOT         | announced | wheeled AMR platform (drive layout not disclosed) + double p | n/a    | n/a                                                                    | HL Mando Wonju steering plant, 1 pilot unit Dec 2025 (MOTIE M.AX program)                                      |
| <b>CLOiD</b> LG 클로이드<br>LG Electronics               | announced | wheeled autonomous base, low center of gravity               | n/a    | n/a (not disclosed); USD 15,000-25,000 industry estimate               | CES 2026 live demos (fridge retrieval, oven loading, laundry folding); pilot production Jul 2026; no customers |
| <b>Billy</b> 빌리<br>Neubility                         | announced | wheeled (4-wheel vehicle base with cargo compartment)        | n/a    | n/a                                                                    | none yet (concept stage)                                                                                       |
| <b>EIR</b> 에이르<br>Neuromeka                          | announced | wheeled cart-based mobile base (explicitly no legs; installa | n/a    | n/a                                                                    | assembly verification on electric-motor production lines (cable insertion, precision assembly); field trials w |
| <b>DEUX</b> 듀스 (Deuce)<br>XYZ Inc.                   | announced | wheeled (swerve drive, 360° omnidirectional; fixed-base vari | 5.5    | Mobile DEUX USD 39,900; fixed DEUX USD 29,900; Glove X USD 3,900 (pre- | Live demo at LoungeX Seongsu store 2026-07-27 (packing, tidying); no external customers yet; Series B KRW 13B  |
| <b>Friday</b> Friday<br>Holiday Robotics             | prototype | wheeled mobile base (enclosed, compact two-section chassis)  | n/a    | ~USD 200,000 listed; mass-production target ~KRW 100M (~\$72k)         | no named customers; Series A ~KRW 155bn (Jul 2026); manipulation demo videos                                   |
| <b>ILBOT</b> 아이엘봇<br>I.L                             | prototype | wheeled (legless mobile base)                                | n/a    | n/a                                                                    | 2 units in mass production at group affiliate IL Mobility (automotive injection-molding part sorting), from Fe |

| Robot / Company                                       | Status       | Base                                                         | kg/arm | Price       | Deployment evidence                                                                                   |
|-------------------------------------------------------|--------------|--------------------------------------------------------------|--------|-------------|-------------------------------------------------------------------------------------------------------|
| <b>TR Works</b> TR웍스<br>T-Robotics                    | prototype    | wheeled (company's own AMR mounted beneath humanoid tor      | n/a    | n/a         | none yet; first public showing AW 2026                                                                |
| <b>ALLEX</b> 알렉스<br>WIRobotics                        | prototype    | pedestal (upper-body system); wheeled mobile config and bipe | 3      | USD 180,000 | none disclosed; RLWRLD physical-AI partnership                                                        |
| <b>M-Hubo</b><br>KAIST Hubo Lab                       | research     | wheeled (omnidirectional base)                               | 10     | n/a         | public exhibition demo: beverage fetching at 24% human speed, 90% success controlled / 80% dynamic    |
| <b>AMBIDEX</b> 앰비덱스<br>NAVER LABS                     | research     | fixed pedestal/stand (upper-body dual-arm research robot, no | 3      | n/a         | cleans Rookie delivery-robot fleet in NAVER 1784 HQ; IROS 2018 challenge winner; no external installs |
| <b>DAL-e</b> 달이<br>Hyundai Motor Group (Robotics Lab) | discontinued | wheeled, 4 omnidirectional wheels                            | 0      | n/a         | pilot at Hyundai showroom, southern Seoul, from 2021-01-25; no broad rollout evidenced                |
| <b>CIROS</b><br>KIST                                  | discontinued | wheeled (omnidirectional)                                    | n/a    | n/a         | exhibition demos only (Robot World 2012 salad-making)                                                 |

## 11 Regional landscape — Rest of world

**22 documented robots.** Israel: Unlimited Robotics' Gary (~10 units in a Philadelphia hospital plus Israeli MoH deployments) — and Mobileye's \$900M acquisition of Mentee (bipedal line; the rumored wheeled variant does not exist) shows the exit ceiling. India: from Invento's wound-down Mitra to Genrobotics' 200-kg-payload tracked GenBot, Perceptyne's assembly torsos and ISRO's space-bound Vyommitra (Gaganyaan G-1, now Q3 2027). Russia: Promobot's V.4 service fleet (800+ units claimed) persists, but its Robo-C android line is discontinued and the FCC action closes the US. Turkey's AKINROBOTICS, Saudi Arabia's QSS, Nepal's Paaila and Vietnam's defunct TOPIO show the long tail. The open-source movement matters strategically: XLeRobot (~\$660, 5.4k GitHub stars), BamBot (~\$300), InMoov and hobbyist builds like Nyro (differential-gear joints, QDD motors) are training a generation on semi-humanoid architecture — tomorrow's engineers and tomorrow's expectations.

| Robot / Company                                                | Status    | Base                                                          | kg/arm | Price                                                                   | Deployment evidence                                                                                            |
|----------------------------------------------------------------|-----------|---------------------------------------------------------------|--------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>ADA-7</b> ADA-7 Sosyal Robot<br>AKINROBOTICS (AKINSOFT)     | shipping  | wheeled (concealed base), 360-degree rotation on the spot     | n/a    | n/a (sale and rental via akinoid.com, prices on request)                | Mini ADA sibling on duty at Istanbul Airport (info/wayfinding); ADA family at Istanbul Robot Museum; fairs/TV/ |
| <b>Abi</b><br>Andromeda Robotics                               | shipping  | wheeled                                                       | n/a    | ~USD 25,000                                                             | Australian aged-care pilots (vendor-reported)                                                                  |
| <b>Sophia</b> Sophia (索菲亞)<br>Hanson Robotics                  | shipping  | wheeled tripod (passive standard; motorized RC or autonomous) | 0.5    | n/a (never officially published); third-party estimates USD 30k-100k+;  | appearances in 65+ countries; UN/ITU AI for Good fixture incl. 'Sophia and Robert' (Jul 2026); R&              |
| <b>BamBot B0</b><br>Open-source project (Tim Qian)             | shipping  | wheeled (small differential base, 3D-printed)                 | n/a    | ~USD 300 in parts                                                       | ~956 GitHub stars / 93 forks; community builds; viral demo 2025-03                                             |
| <b>Ginger (Nepal)</b><br>Paaila Technology                     | shipping  | wheeled (concealed base under humanoid torso)                 | n/a    | n/a                                                                     | 3 Ginger + 2 Ferry robots serving at Naulo restaurant since 2018; Pari (2 units/generations) at Nepal SBI Bank |
| <b>Promobot V.4</b><br>Промобот V.4<br>Promobot                | shipping  | wheeled (2-wheel diff-drive + support casters)                | 0      | from USD 25,000; ~\$2,000/day rental                                    | 800+ units / 40+ countries (vendor claims 43 countries); Perm MFC civil-servant robot 30k+ inquiries; Kuwait O |
| <b>Gary</b> Gary<br>Unlimited Robotics                         | shipping  | wheeled (four-wheel skirted base)                             | n/a    | ~USD 30,000 + monthly fee (hospital model); 2021 consumer plan \$99 wai | ~12 units at Rabin + Shamir medical centers since early 2024; Israel MoH supply agreement; ~10 units at a Phil |
| <b>TARA GEN1</b><br>iHub Robotics (IHUB Research and Robotics) | shipping  | wheeled circular base under skirt cowlings                    | n/a    | USD 13,998                                                              | exports to UAE/Saudi Arabia reported, no unit counts or named customers; listing 'Not Verified'                |
| <b>Spaceo M1 / Spaceo Pro</b><br>Muks Robotics                 | announced | wheeled box base (M1); wheeled chassis with adjustable height | 10     | M1 from ~USD 11,000; Pro USD 60,000                                     | none found; no named customers or pilots                                                                       |
| <b>PR-34D</b><br>Perceptyne                                    | announced | stationary pedestal/pillar mount (no mobility by design)      | 6      | ~USD 120,000                                                            | pilots with unnamed Indian automotive/electronics manufacturers; \$3M seed Oct 2024 (Endiya, Yali, Whiteboard) |

| Robot / Company                                                                                        | Status       | Base                                                            | kg/arm | Price                                                      | Deployment evidence                                                                                            |
|--------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------|--------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Bimanual</b><br>Svaya Robotics                                                                      | announced    | stationary pedestal<br>(mountable on third-party mobile bases a | 10     | ~USD 90,000                                                | no named customers/units; company credibility via DRDO quadruped+exoskeleton demonstrators and SR-L cobot line |
| <b>GenBot</b><br>Genrobotics<br>(Genrobotic Innovations)                                               | prototype    | tracked - rubberized quad-track chassis with adaptive multi-    | n/a    | n/a                                                        | none for GenBot; company's Bandicoot sewer robots deployed in 22 Indian states + 4 countries                   |
| <b>Vyommitra</b> व्योममित्र<br>ISRO (Vikram Sarabhai Space Centre)                                     | prototype    | none - legless seated half-humanoid (head + torso + 2 arms),    | n/a    | n/a                                                        | flight integration for Gaganyaan-1 since 2026-04-28; G1 launch targeted Q3 2027 (risk 2028; slipped from H2 20 |
| <b>Sara</b> سارة<br>QSS AI & Robots                                                                    | prototype    | concealed legless lower body (estimated wheeled pedestal)       | n/a    | n/a                                                        | event appearances: LEAP, DeepFest, AI for Good, Arabs Got Talent; QSS plan for up to 10,000 humanoids in 5 yea |
| <b>TM Xplore 1</b><br>Techman Robot                                                                    | prototype    | wheeled                                                         | n/a    | ~USD 25,000 (indicative)                                   | none public                                                                                                    |
| <b>Nadine</b><br>NTU Singapore (IMI); hardware by Kokoro (Japan); research continued at MIRALab Geneva | research     | seated/stationary android (no locomotion)                       | n/a    | n/a                                                        | ArtScience Museum 2017 (>100k interactions); AIA Singapore customer service; Bright Hill Evergreen Home eld    |
| <b>KIKI</b> КИКИ<br>Alfa Robotics                                                                      | discontinued | wheeled (concealed skirt-style mobile base)                     | n/a    | n/a                                                        | promo/event deployments in Russia claimed; no counts or named customers found                                  |
| <b>KP-BOT</b><br>Asimov Robotics Pvt Ltd                                                               | discontinued | wheeled (concealed base under skirt-style lower body)           | n/a    | INR 21.24 lakh (~USD 25,500)                               | 1 unit at Kerala Police HQ 2019--2021; withdrawn to Cyberdome for training/research by 2025                    |
| <b>Mitra</b> मित्र<br>Invento Robotics                                                                 | discontinued | wheeled (concealed base, layout not disclosed)                  | 0      | n/a (est. ~USD 10-15K class per period press)              | GES 2017 Modi/Ivanka greeting; banks/hotels/malls/cinemas/airports 2017-21; COVID patient telepresence at Yath |
| <b>Robo-C / Robo-C2</b><br>Робо-С / Робо-С2<br>Promobot                                                | discontinued | stationary pedestal/seated android (no locomotion)              | 1      | historically ~USD 20,000-50,000 depending on customization | small numbers since 2019: service halls, museums, expos, celebrity doubles; Perm State Medical University pati |
| <b>TOPIO Dio</b><br>TOSY Robotics JSC                                                                  | discontinued | wheeled (3-wheel omnidirectional platform)                      | n/a    | n/a                                                        | trade-show demos only (AUTOMATICA 2010); no customers documented                                               |

## 12 Technology trends

### Foundation models made the category investable

Vision-language-action (VLA) models are the demand engine: Physical Intelligence's  $\pi 0/\pi 0.5$  run long-horizon household tasks on *wheeled* mobile manipulators (\$5.6B valuation, ~\$1B raise in talks); NVIDIA's GR00T, Google's Gemini Robotics 2 (whole-body control, adapts to new hardware with <200 examples), AgiBot's GO-2, Galbot's GraspVLA (billion-frame synthetic pretraining) and Figure's Helix define the stack. Consequence: hardware is increasingly judged as a **data-collection and policy-execution vehicle**. AgiBot World's million-trajectory dataset was collected on wheeled G2s; Sunday's 2,000 paid "memory developers" wear gloves whose geometry its robot hand mirrors; China operates 40+ state-funded training centers; 1X's EVE security fleet trained NEO. Teleoperation is not a stopgap but a revenue-generating data business (Reflex, ugo, Beomni) — Interact Analysis expects deployments to remain "predominantly semi-autonomous" near-term.

### Hardware convergence

**The 2026 median semi-humanoid:** two 7-DoF arms; 20–40 total DoF; 3–5 kg per-arm payload (10–20 kg for logistics classes; outliers: Dexterity 60 kg, Duatic 30 kg peak, Realman 5 kg official); 1.3–1.75 m height, often with a lift column or folding torso (floor-to-2 m+ work envelopes: Galbot 2.4 m, AI<sup>2</sup> 0–2.4 m, Dexmate 0.66–2.2 m); omnidirectional wheels at ~1.5–2 m/s; 5–8+ h runtime with hot-swap batteries; 3D lidar + multi-RGB-D + wrist F/T sensing; NVIDIA Jetson Orin (now Thor: Richtech Dex, Dexmate) as de-facto compute; ROS 2 plus a proprietary autonomy/VLA layer. Actuation splits between QDD/backdrivable philosophies (Toyota ELEY, Duatic DynaArm, Spirit AI full force control) and harmonic/planetary precision claims ( $\pm 0.03$ –0.1 mm: JAKA, Elite, iSageBot).

### Hands and end-effectors

Parallel grippers still dominate deployments; dexterous hands are differentiating fast: Chinese five-finger hands now start under \$1,400 (Inspire ubiquitous; the market grew ~150% YoY into a "mass-production race"), Western options run to Shadow's ~\$300k; in-house designs (Pudu DH11, XYZ's 7-DoF 3-finger + Glove X, Sanctuary's hydraulic 21-DoF, PaXini's ITPU tactile arrays) anchor product identity. Tool-changers and media at flange remain rare disclosures — an open differentiation axis for industrial-grade entrants. Notably, several "humanoid" vendors ship off-the-shelf EEs (1X EVE used a Robotiq 3-finger; OpenDroids integrates RealMan hands).

## 13 Pricing & business models

| Band      | Examples (2026 street/list prices)                                                                                                                                                                                                                                                                            | Model                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <\$5k     | Unitree R1-D ¥26,900 (~\$4.3k); open kits (XLeRobot \$579–660, BamBot ~\$300)                                                                                                                                                                                                                                 | Entry/education; ecosystem seeding                            |
| \$5–30k   | Weave Isaac \$7,999 or \$449/mo; Almond Axol from \$7,999; Atribot T1 ~\$14k China / \$40,950 via Western resellers; Elephant Mercury X1 \$15,999; SwitchBot Onero H1 target ~\$1–1.5k (pre-production); WorkFar tiers \$10–60k; XYZ DEUX \$29.9–39.9k; Techman ~\$25k est.                                   | Home subscriptions; research/edu; prosumer                    |
| \$30–100k | Deft Simba \$34,900; igus Iggy Rob €47,999; AgiBot A2 Lite \$44.6k; A2-W ¥400–800k insider; Atribot S1 ~\$69k; Reachy 2 ~\$70k; UMA Northstar ~\$70k; Noble \$75k; Rainbow RB-Y1 \$80k research; Dexmate Vega \$89,999; Genesis Eno ~\$90k; Galbot G1 ~\$91–97k; Keenon XMAN ~\$100k; PAL TIAGo Pro ~\$85–95k | Industrial/research purchase; the segment's center of gravity |

| Band    | Examples (2026 street/list prices)                                                                                                                     | Model                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| >\$100k | Rainbow RB-Y1 commercial \$120k; Richtech ADAM \$180k; Ameca ~\$250k; CASBOT 02-class ¥328.8k; CloudMinds Ginger ¥698k; research heritage (PR2 \$400k) | Premium/entertainment/legacy                                     |
| RaaS    | ADAM \$3,500/robot/mo (25-unit 60-mo contract); AgiBot A2 ~\$20k+/mo; delivery-class \$300–800/mo; Eden \$10/hour; home \$449–499/mo (Weave, 1X NEO)   | Labor-substitution pricing; dominant new-entrant model in the US |

Note the China-vs-export gap: the same robot often costs 2–3× more through Western resellers (Atribot T1: ~\$14k domestic vs \$40,950 US reseller list) — tariffs, support and channel margins that an EU-based vendor does not pay locally. Pattern: Chinese street prices cluster at \$45–100k industrial (falling — some wheeled models below RMB 100k), Western/Korean at \$80–180k, and a new sub-\$15k consumer/edu front opened in 2025–26. Maintenance contracts (Deft: \$30k/yr on a \$40k robot) and software/support subscriptions (Galbot ~\$27k/robot/yr) shift lifetime value off the sticker price. History's warning: Pepper's ~\$9k consumer bundle lost money at ~27,000 units and the business died; unit economics, not unit counts, decide survival.

## 14 Safety, standards & regulation

- **ISO 13482** (personal care robots — mobile servant class fits semi-humanoids): 2014 edition current; revision (retitled toward "service robots", broadened to professional applications, FDIS stage mid-2025) expected imminently.
- **ISO 10218-1/-2:2025** published: industrial robot safety overhauled with functional safety, cybersecurity and collaborative requirements; TS 15066 folded in. **ANSI/RIA R15.08** covers industrial mobile robots with mounted manipulators; **UL 3300** (service robots in public spaces) joined OSHA's NRTL list Dec 2025.
- **The wheeled certification advantage:** statically stable platforms certify under all of the above *today*; dynamically balancing robots await **ISO 25785-1** (committee draft; expected 2026–27). IEEE's humanoid-standards pathway study (2025) puts ratified humanoid standards 18–36 months out.
- **EU:** Machinery Regulation 2023/1230 applies **20 January 2027** (no grace period): self-evolving-behaviour assessment, cybersecurity of safety functions, digital documentation — layered with the AI Act. Burden for laggards; moat for compliance-by-design vendors. TÜV-grade certification of a dual-arm mobile platform is an unclaimed European first.
- **USA — the 2026 discontinuity:** FCC Public Notice **DA 26-786** (28 July 2026) added "foreign-produced advanced robotic devices" to the Secure Networks Act Covered List: any mobile ground robot >4.4 lbs with environmental sensing, ≥200 kbps connectivity and AI autonomy loses FCC equipment authorization for new models unless granted a Department-of-War conditional approval. Fixed industrial robots are exempt; wheeled/tracked semi-humanoids are squarely in scope. Practical effects: Chinese vendors lose the US; Korean/Japanese/European vendors need approvals or US manufacturing ("domestic end product" per 48 CFR 25.101(a)); US integrators re-source components.
- **China:** MIIT stood up a humanoid/embodied-AI standardization committee (Nov 2025; Unitree and AgiBot founders as deputy chairs) and unveiled the first national humanoid standard system in March 2026; existing GB/T service-robot safety standards apply meanwhile. Shanghai issued the first humanoid governance guidelines (2024).

**Observed reality:** among 180 main-matrix robots, published robot-safety certifications are rare to absent — including for market leaders. Oversonic's "industrial + healthcare certified" claim resolves to CE-machinery + ISO 27001. A vendor that can put ISO 13482 / MR-2027-ready CE on a dual-arm mobile platform holds a genuinely scarce asset.

## 15 Deployment evidence: who is really working

| Deployment class          | Evidence (confidence)                                                                                                                                                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Historic scale            | Pepper: ~27,000 produced / ~17,000 sold, worldwide retail & reception (third-party) — still unmatched                                                                                                                                                                                                    |
| Factory production shifts | AgiBot A2-W at Fulin Precision (~100-unit order; 4 robots moved 800+ boxes/3 h) and G2 at Longcheer (310 units/h tablet line) (third-party); Walden W0.5 in 8-h shifts at a Toyota plant since Feb 2026 (vendor/third-party); Midea MIRO U in its Wuxi plant (vendor); Zoomlion internal fleets (vendor) |
| Retail/pharmacy autonomy  | Galbot: 10+ licensed Beijing robot pharmacies, ~170 kiosks in 40+ cities, 99.5% grasp claim over ~5,000 SKUs (vendor-claimed); Haohai "Juliet" pods on Nanjing Road (2,368 orders in first months, third-party)                                                                                          |
| Hospitality/F&B           | Richtech ADAM: Walmart-hosted One Kitchen sites, Vegas café (16k+ drinks), arenas (vendor/third-party); Keenon XMAN-R1 at Shangri-La; Zerith H1 in 20+ hotels (third-party)                                                                                                                              |
| Healthcare/eldercare      | Gary: ~10 units Philadelphia + Israeli MoH (third-party); Nyokkey at Fujita hospitals; Mirokai at AP-HP/Montpellier; Aeolus Ae0 in Japanese eldercare (operators are also investors)                                                                                                                     |

| Deployment class        | Evidence (confidence)                                                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security/infrastructure | 1X EVE via Everon/ADT: est. 150–250 US buildings (estimated); ugo Pro in Japanese buildings (third-party); Jinki Ittai on JR West railways — revenue service since July 2024 (third-party) |
| Logistics pilots        | Reflex at GXO; Rainbow RB-Y1 at Coupang (confirmed); Dexterity Mech truck-loading pilots + Sanmina manufacturing (third-party)                                                             |
| Home (beta)             | Weave Isaac deliveries from Sept 2026 (CA); Sunday Memo beta "by Thanksgiving 2026"; 1X NEO late 2026 — all pre-scale (vendor-claimed)                                                     |

Contrast: flagship bipedal deployments remain single-to-low-double-digit robot counts (Figure/BMW: 2 robots; Agility/GXO: "small fleet"). The wheeled category's deployment lead is roughly two orders of magnitude.

## 16 Competitive assessment for a German entrant

### Where the openings are

- **EU-certified industrial semi-humanoid.** No vendor holds robot-safety-grade certification for a dual-arm mobile platform under the incoming MR 2023/1230 regime. German TÜV culture + the domestic research bench (DLR torque control, KIT ARMAR manipulation, Franka-class compliant arms) make compliance-by-design the natural differentiator against spec-sheet-led Chinese imports.
- **Care & hospital logistics in Europe.** Demographics guarantee demand; the credible platforms are pilots (GARMI has no spin-off; HoLLiE stayed academic; MiPA is late; Mirokai is logistics-only). ISO 13482 + hospital-grade hygiene + Devanthro-style teleop fallback is an assemblable, defensible package.
- **US market via the FCC wall.** A German vendor with US final assembly ("domestic end product") can sell where Chinese rivals now cannot — the single largest regulatory gift to Western robot makers in years. Walden's \$1.1B day-one valuation prices exactly this scarcity.
- **Data flywheel as default architecture.** Ship every unit teleop-capable and log demonstrations under EU-grade data governance ("GDPR-clean embodied-AI data" is itself a product for European enterprise customers wary of Chinese and US clouds).
- **Component strategy.** Schaeffler (actuators; invested across Humanoid UK, Hexagon, reportedly Neura), igus (polymers), Franka/Agile (arms), SICK/IDS (sensing) form a domestic supply chain; the missing integrator role is the opportunity.

### Threat tiers (analyst view)

| Tier   | Actors                                                                                                            | Why                                                                                            |
|--------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Severe | AgiBot, Galbot, UBTech; Keenon/Pudu channels; Unitree price disruption                                            | Scale, capital, data, falling prices; EU entry inevitable (Rokae already exhibits in the West) |
| High   | Toyota ELEAY, Yaskawa(+Tokyo Robotics), Samsung/Rainbow, LG, Hexagon/Schaeffler, Walden                           | Incumbent capital + manufacturing + patient roadmaps                                           |
| Medium | Humanoid UK, Neura, Sunday/Weave/Dexterity/Dyna, Richtech, Oversonic, Enchanted Tools                             | Funded and fast, but each thesis still unproven at scale                                       |
| Watch  | SwitchBot/consumer-price entrants; open-source kits; Genesis-class model labs; Vassar/Piggy/HABIT-type YC stealth | Price-floor and expectation-setting effects                                                    |

### Recommended posture

**Enter as the certified European industrial/care semi-humanoid with a data flywheel.** Concretely: (1) wheeled omnidirectional base + lift torso, dual 7-DoF compliant arms in the 5–10 kg/arm class, tool-changer + media at flange (an undisclosed axis across nearly all competitors); (2) MR-2027/ISO-13482 compliance engineered from day one and certified loudly; (3) launch vertical where ROI is provable within one budget cycle — intralogistics machine-tending or hospital logistics — with RaaS pricing against the \$3.5k/mo ADAM benchmark; (4) teleop + demonstration-data capture in every unit; (5) US strategy via local assembly partner from year one; (6) strategic component alliances (Schaeffler/Franka/igus) rather than full vertical integration. Speed matters: the Korean incumbent wave shows how fast listed industrials can pivot, and Chinese price curves compress the window in which "European + certified + trusted" commands a premium.

## 17 Annex A: Borderline & excluded systems

Documented for completeness; excluded from the main matrix. Full dossiers exist for each in the repository.

| Robot / Company                                               | Base                                                            | Classification note                                                                                                                |
|---------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>GoMate</b><br>GAC Group                                    | variable wheel-leg: 4-wheel mode (~1.4 m) / 2-wheel self-bal    | GAC all-solid-state battery; claimed >80% energy reduction vs comparable humanoids. Third-generation GAC embodied robot. BORDERLIN |
| <b>Stretch 3</b><br>Hello Robot                               | wheeled (differential drive) —<br>BORDERLINE: single-arm, not a | BORDERLINE ENTRY (single arm) — include in borderline annex, not the main matrix. Added at user request as design-philosophy refer |
| <b>AEON</b><br>Hexagon Robotics<br>(Hexagon AB)               | BORDERLINE wheels-on-legs: articulated legs terminating in p    | BORDERLINE morphology (wheels-on-legs) — flag when comparing against pure wheeled semi-humanoids. Strategic angle: metrology giant |
| <b>EMIEW3</b><br>Hitachi                                      | wheeled: drive wheels + small casters per foot; EMIEW lineag    | Borderline inclusion: no manipulation capability. EMIEW = Excellent Mobility and Interactive Existence as Workmate. Original 2005  |
| <b>MenteeBot V3.0</b><br>Mentee Robotics (Mentee by Mobileye) | bipedal (WHEELED VARIANT UNVERIFIED - discovery claim likely    | Folder named menteebot-wheeled per assignment, but NO wheeled variant exists per all checked sources — discovery's 'modular biped  |
| <b>L1</b><br>Moon Dynamics                                    | two-wheel dynamic-balancing wheel-LEG hybrid (BORDERLINE sco    | BORDERLINE scope: wheel-leg (wheeled-bipedal) robot, included as boundary datapoint like Hexagon AEON / GAC GoMate. Company extrem |
| <b>Moby</b><br>Noble Machines (ex-Under Control Robotics)     | BIPEDAL/legged (NOT wheeled - contradicts discovery classifi    | SCOPE: bipedal legged humanoid, not a wheeled semi-humanoid - discovery note '9kg lift, wheeled' was wrong on both counts (payload |
| <b>Bot Handy</b><br>Samsung Research<br>(Samsung Electronics) | wheeled cylindrical base (drive type not disclosed); single-    | Borderline: single-arm concept, included for Samsung-strategy context. Samsung's real semi-humanoid position runs via 35% subsidia |
| <b>The Five (XiaoWu)</b><br>Tencent Robotics X                | wheel-leg hybrid: four straight telescoping legs with wheels    | BORDERLINE for study scope: wheel-leg hybrid (4 straight telescoping legs with wheels), not a wheeled-base semi-humanoid. Fifth-ge |
| <b>Human Support Robot (HSR)</b><br>Toyota                    | wheeled (omnidirectional) —<br>BORDERLINE: single-arm, not a se | BORDERLINE ENTRY (single arm) — borderline annex, not main matrix. Added at user request as the historic home-assistance archetype |
| <b>Calvin-40</b><br>Wandercraft                               | BIPEDAL (two legs, headless) —<br>EXCLUDED from semi-humanoid s | EXCLUDED-as-biped: kept in study for Renault deployment context. Survey-sweep 'wheeled' claim was wrong; startup-sweep 'biped' cla |

## 18 Annex B: Research repository guide

This study compresses a research repository delivered alongside it:

- **companies/<company>/company.md** — company profile: ownership, funding, strategy, sources.
- **companies/<company>/<robot>/robot.md** — full dossier: morphology, locomotion, manipulation, sensing, actuation, compute/software, safety, deployment evidence, analyst assessment, numbered sources.
- **.../specs.json** — machine-readable spec sheet; every field carries value + confidence (vendor-claimed / third-party / estimated) + source URL. These files generate the xlsx.
- **.../images/** — downloaded imagery, each visually verified; images.md maps every file to its source page, direct URL, caption and licence note.

- **semi\_humanoid\_market\_2026.xlsx** — Overview, full 34-column spec matrix, per-region sheets, Borderline & excluded, ~3,400-row per-field source register, confidence colour legend.

Compiled 2 August 2026. Sources were accessed August 2026; the per-field register in the xlsx is authoritative for citations. Confidence discipline: vendor-claimed figures are marketing until deployed; third-party validation is noted wherever it exists; conflicts are preserved, not averaged.